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A Multimodal System for Automated Depression Detection Using Deep Learning and Audio-Visual Signal Analysis

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**A dissertation submitted in partial fulfillment of the requirements for the degree
of Digilians 9 Months Diploma in Applied AI and Data Analytics**

Supervised By

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Abstract

Major Depressive Disorder (MDD) is one of the leading causes of disability worldwide, affecting more than 280 million people and imposing substantial social and economic burdens on healthcare systems. Current depression assessment relies largely on subjective clinical instruments such as the Patient Health Questionnaire-9 (PHQ-9) and Hamilton Depression Rating Scale (HAM-D), which face reporting bias and scalability challenges. Although automated approaches have shown promise, many existing systems rely on a single modality, either speech or facial expressions, which may overlook complementary affective information.

This study presents MindSync, a feature-level multimodal deep learning framework for automated depression-indicator detection from synchronized audio-visual recordings. The framework integrates independently trained audio and video sub-networks. The audio branch processes 40-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) using four one-dimensional convolutional blocks, two Bidirectional Long Short-Term Memory (BiLSTM) layers, and a Bidirectional SimpleRNN layer. The video branch employs a pretrained R3D-18 three-dimensional convolutional network trained on Kinetics-400, enhanced with additional Conv3D layers for task-specific spatiotemporal facial feature extraction. The resulting modality-specific representations are combined with their Softmax probability distributions to construct a 568-dimensional hybrid feature vector, classified using a Multi-Layer Perceptron (MLP) fusion network optimized with AdamW and StepLR scheduling.

MindSync is trained and evaluated on the RAVDESS dataset, containing 2,452 synchronized audio-visual recordings across eight emotion categories performed by 24 professional actors. To address class imbalance, Synthetic Minority Over-sampling Technique (SMOTE) is applied exclusively to the training partition within the MFCC feature space, preventing data leakage into the evaluation set.

A transparent rule-based Mental Health Inference Engine (MHIE) maps the eight emotion probabilities into four affective indicators: possible depression, anxiety, stress, and emotional blunting, representing emotional-pattern analysis rather than a clinical diagnosis.

Experimental results demonstrate accuracies of 85.55%, 96.54%, and 96.74% for the audio-only, video-only, and multimodal systems, respectively, with corresponding macro AUC-ROC values of 98.60%, 99.46%, and 99.38%. The superior performance of the multimodal system demonstrates that integrating complementary audio-visual representations provides a richer, more discriminative representation than either modality alone. Future work will focus on domain adaptation using clinical datasets such as DAIC-WOZ, integration of physiological signals, and prospective validation against clinician-administered depression severity scales.

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List of Abbreviations

Abbreviations

1D-CNN	One-Dimensional Convolutional Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
AdamW	Adam Optimizer with Decoupled Weight Decay (Loshchilov & Hutter, 2019)
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
AVEC	Audio/Visual Emotion Challenge
BDI	Beck Depression Inventory
BERT	Bidirectional Encoder Representations from Transformers
BiLSTM	Bidirectional Long Short-Term Memory
CCE	Categorical Cross-Entropy (Loss Function)
CNN	Convolutional Neural Network
DAIC-WOZ	Distress Analysis Interview Corpus – Wizard of Oz
DCT	Discrete Cosine Transform
DCRF	Deep Conditional Random Field
DLS	Depression Likelihood Score
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
EDA	Exploratory Data Analysis
F0	Fundamental Frequency (Pitch)

F1-Score	Harmonic Mean of Precision and Recall
FACS	Facial Action Coding System
FC	Fully Connected (Layer)
FER	Facial Emotion Recognition
GAP	Global Average Pooling
GDPR	General Data Protection Regulation (EU, 2018)
GRU	Gated Recurrent Unit
HAM-D	Hamilton Depression Rating Scale
HIPAA	Health Insurance Portability and Accountability Act (US, 1996)
HNR	Harmonic-to-Noise Ratio
IEMOCAP	Interactive Emotional Dyadic Motion Capture Database
INT8	8-bit Integer Quantization (for model compression)
LLM	Large Language Model
LSTM	Long Short-Term Memory
MADRS	Montgomery–Åsberg Depression Rating Scale
MDD	Major Depressive Disorder
MFCC	Mel-Frequency Cepstral Coefficient
MHIE	Mental Health Inference Engine
ML	Machine Learning
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error

ONNX	Open Neural Network Exchange (model interchange format)
PHQ-9	Patient Health Questionnaire-9
R3D-18	3D Residual Neural Network with 18 Layers (Tran et al., 2018)
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
ReLU	Rectified Linear Unit
ResNet	Residual Neural Network (He et al., 2016)
RGB	Red, Green, Blue (colour channel format)
RNN	Recurrent Neural Network
SER	Speech Emotion Recognition
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SSL	Self-Supervised Learning
STFT	Short-Time Fourier Transform
SVM	Support Vector Machine
ViT	Vision Transformer
WHO	World Health Organization
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once (real-time object detection architecture)
YOLOv8	You Only Look Once – Version 8 (Ultralytics, 2023)
ZCR	Zero-Crossing Rate

Chapter 1

Introduction

1.1 Overview

Mental health disorders constitute one of the most consequential and, paradoxically, one of the most insufficiently addressed challenges confronting contemporary healthcare systems worldwide. Among these conditions, Major Depressive Disorder (MDD) stands out as one of the most prevalent, disabling, and economically burdensome disorders, affecting individuals across all demographics, cultural, and socioeconomic strata. Clinically, MDD is characterized by a constellation of symptoms including persistent low mood, anhedonia, cognitive impairment, psychomotor retardation, disrupted sleep architecture, appetite disturbances, and, in its most severe manifestations, suicidal ideation.

Despite the existence of effective pharmacological and psychotherapeutic interventions, a substantial proportion of individuals living with depression remain either undiagnosed or inadequately treated. This persistent diagnostic gap is primarily attributable to entrenched social stigma surrounding mental illness, a chronic global shortage of trained mental health professionals, and the inherent methodological limitations of conventional clinical assessment, which typically relies on subjective clinician judgment and episodic, rather than continuous, observation.

A critical premise underpinning this work is that depression leaves measurable and observable traces across multiple physiological and behavioral modalities, rendering it amenable to computational analysis. At the level of facial behavior, depressive states are consistently associated with a reduced frequency and intensity of spontaneous smiling, diminished overall facial expressivity, slower onset and offset dynamics of emotional expressions, and an increased prevalence of neutral or negative facial configurations relative to non-depressed individuals.

At the level of speech acoustics, depression similarly produces a characteristic and measurable set of alterations. These include reduced variability in fundamental frequency (F0), flattened prosodic contours, slower articulation rate, an increased frequency and duration of pauses, diminished vocal energy, and altered harmonic-to-noise ratios. Such alterations can be quantitatively captured through established spectral and cepstral descriptors,

most notably Mel-Frequency Cepstral Coefficients (MFCCs) and chroma-based representations, both of which provide meaningful, low-dimensional inputs suitable for downstream computational modeling.

Taken together, these multimodal behavioral and physiological indicators constitute the empirical foundation upon which automated depression-detection systems can be built. Artificial Intelligence, and deep learning in particular, has fundamentally transformed the landscape of clinical and biomedical data analysis over the past decade. Architectures based on Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and, more recently, Transformer-based models have repeatedly demonstrated state-of-the-art performance across a broad spectrum of healthcare applications, ranging from medical imaging diagnostics to physiological-signal interpretation.

Within the specific sub-domain of Speech Emotion Recognition (SER), recent studies report remarkable classification performance using deep architectures. Lightweight one-dimensional CNN models employing multi-feature fusion have achieved an accuracy of 91.9% on the RAVDESS benchmark dataset [1]. Majority-voting ensembles built upon large self-supervised transformer backbones have reached 96.88% accuracy [2], while hybrid Deep Conditional Random Field – Bidirectional LSTM (DCRF-BiLSTM) frameworks have reported accuracies as high as 97.83% [3]. These results collectively underscore the effectiveness of contemporary deep learning techniques in extracting and modeling the complex, time-varying emotional patterns embedded within speech signals.

The system proposed in this thesis, referred to throughout as MindSync, adopts a feature-level fusion strategy in which deep feature representations are extracted independently from the video and audio modalities and subsequently concatenated into a unified joint embedding space prior to binary classification. This design choice allows the model to exploit complementary cross-modal information while remaining computationally tractable, and it forms the architectural backbone that is elaborated in detail throughout the remainder of this thesis.

From an implementation standpoint, the Speech Emotion Recognition pipeline begins with the preprocessing of raw audio input through established vocal feature-extraction techniques, namely Mel-Spectrogram analysis, MFCC computation, and Chroma feature extraction. To improve robustness against real-world acoustic variability, data augmentation strategies such as Gaussian noise injection and time-shifting are systematically applied during training. The resulting processed features are subsequently propagated through a multi-layer convolutional neural network, which learns hierarchical, increasingly abstract representations and ultimately produces the final classification output.

The dataset selected for experimental evaluation is the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS), comprising 7,356 audio-visual recordings performed by 24 professionally trained actors expressing eight distinct emotional states under controlled recording conditions. For the specific purposes of this project, the original eight-class emotion taxonomy is re-mapped onto a binary classification scheme: recordings corresponding to emotional states that are clinically associated with depressive affect, namely sadness, fear, and related negative-valence, low-arousal expressions, are labeled as “Depressed,” whereas recordings reflecting neutral or positive emotional states are labeled as “Normal.” This re-labeling transformation enables the underlying RAVDESS corpus, which was not originally collected for clinical purposes, to be repurposed in a principled manner for the specific task of automated depression-indicator detection.

The high-level architecture of the MindSync system, illustrating the dual-modality input design and the flow from independent feature extraction through fusion to final classification, is depicted schematically in Figure 1.1 below. Audio signals are processed through a dedicated feature extraction branch that captures MFCC, pitch, energy, and spectral descriptors, while visual signals are handled by a parallel branch that extracts CNN embeddings, facial landmarks, and expression dynamics. The outputs of both branches are subsequently combined within a fusion layer employing early, late, or hybrid fusion strategies, and the resulting joint representation is passed to a classification module that produces the final depression-detection output.

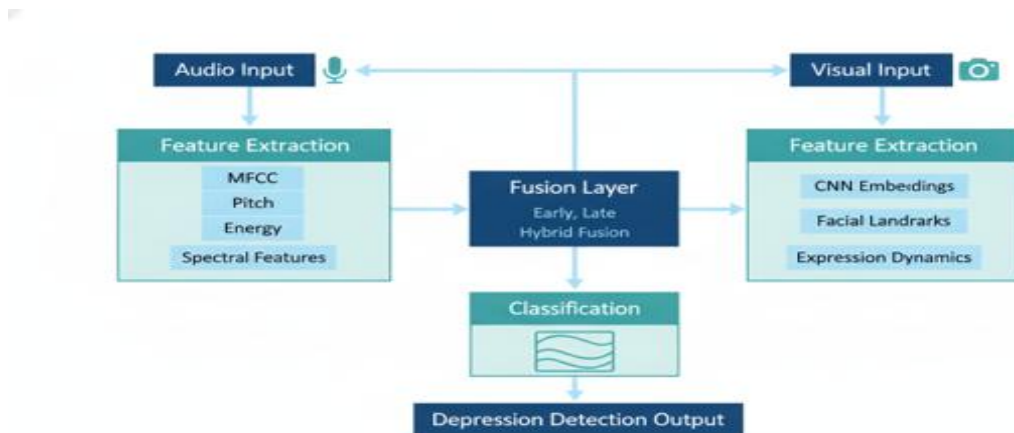


Figure 1.1: High-Level Architecture of the Proposed MindSync Multimodal Depression Detection System.

1.2 Motivation

1.2.1 Global Prevalence and Social Consequences of Depression

Depression constitutes one of the most significant public health challenges of the twenty-first century. According to the World Health Organization (WHO), depressive disorders affect hundreds of millions of

individuals worldwide and represent one of the leading causes of years lived with disability across all age groups [4]. Furthermore, projections derived from the Global Burden of Disease study indicate that depression is set to become the foremost contributor to overall disease burden within the coming decade, surpassing cardiovascular and respiratory diseases in terms of cumulative functional impairment. It is important to emphasize that these figures are likely conservative underestimates, given the pervasive under-diagnosis and cultural barriers to self-reporting that characterize many low- and middle-income regions.

Beyond its clinical dimension, depression exerts profound and far-reaching social consequences. It disrupts interpersonal relationships, diminishes academic and occupational performance, and is closely associated with elevated risks of substance abuse and suicide. The cumulative societal cost of untreated depression therefore extends well beyond the individual patient, permeating families, educational institutions, and workplaces alike.

1.2.2 Economic Burden and Healthcare System Impact

The economic impact of depression is both substantial and multidimensional in nature. Direct healthcare costs, encompassing psychiatric consultations, pharmacological treatment regimens, hospitalization, and crisis intervention services, represent only a fraction of the total economic burden imposed by the disorder. Indirect costs, including reduced workforce participation, chronic absenteeism, diminished productivity, and premature mortality, significantly exceed these direct expenditures in aggregate terms. Global estimates suggest that the combined economic losses attributable to depression and anxiety disorders amount to hundreds of billions of US dollars annually, with several authoritative studies reporting cumulative figures exceeding one trillion USD in lost productivity alone [5].

These economic pressures place an increasingly unsustainable strain on national healthcare systems, particularly in regions where mental health infrastructure is already under-resourced. Consequently, there exists a pressing and well-documented need for cost-effective, scalable screening mechanisms capable of identifying at-risk individuals at earlier stages of disease progression, thereby reducing the long-term economic and clinical burden associated with delayed diagnosis.

1.2.3 Limitations of Traditional Clinical Assessment

Current clinical approaches to diagnosing depression rely predominantly on structured interviews and standardized self-report instruments, including the Hamilton Depression Rating Scale (HAM-D), the Patient Health Questionnaire-9 (PHQ-9), the Beck Depression Inventory (BDI), and the Montgomery-Aberg Depression

Rating Scale (MADRS). Although these instruments are clinically validated and widely deployed in both research and practice, they remain subject to several fundamental and well-recognized limitations.

These limitations include an inherent degree of subjectivity influenced by individual clinician judgment and interpretive bias; a reliance on episodic assessments that capture only discrete temporal snapshots of an inherently dynamic and fluctuating condition; a chronic global shortage of trained mental health professionals, particularly acute in rural and low-resource settings; and the persistent social stigma that continues to discourage many individuals from proactively seeking professional help. Moreover, patient self-reporting biases, social desirability effects, and cross-cultural variations in the expression and interpretation of emotional symptoms further compromise the consistency and diagnostic accuracy of these traditional instruments [6].

1.2.4 The Case for Automated and Scalable Screening

Automated computational systems offer a fundamentally different and complementary approach to depression screening that directly addresses many of the limitations outlined above. By leveraging passively collected or readily accessible audiovisual data, such as short video recordings of natural speech, these systems can deliver objective, reproducible, and highly scalable assessments without necessitating continuous, real-time clinician involvement.

It is essential to underscore that the intended role of artificial intelligence within this clinical context is not to replace trained healthcare professionals, but rather to augment and extend their diagnostic capabilities. Specifically, such systems are envisioned as initial screening tools, capable of efficiently flagging individuals who may benefit from, and should be referred to, further comprehensive clinical evaluation. This human-in-the-loop philosophy is central to the responsible and ethically sound deployment of AI-assisted mental health technologies.

1.2.5 Advantages of Multimodal Approaches and Feature-Level Fusion

Unimodal depression-detection methods, whether based exclusively on audio or exclusively on video data, are inherently constrained by their reliance on a single source of information, rendering them particularly vulnerable to modality-specific noise, sensor degradation, and contextual ambiguity. In contrast, multimodal approaches integrate complementary information drawn from multiple sensory channels, most notably facial expressions and vocal characteristics, thereby substantially improving overall robustness, reliability, and discriminative power.

Feature-level fusion, the specific multimodal strategy adopted in this thesis, combines deep feature representations extracted from distinct modalities prior to the final classification stage. This approach enables the resulting model to capture rich cross-modal interactions and correlations that may encode depressive patterns considerably more effectively than any single modality could achieve in isolation. Empirical studies in the broader effective-computing literature have consistently demonstrated that multimodal fusion strategies can improve overall classification accuracy by more than fifteen percentage points relative to comparable audio-only baseline approaches [7], providing strong empirical motivation for the system design pursued in this project.

1.3 Challenges

The development of a robust, generalizable, and clinically meaningful automated depression-detection system based on synchronized audiovisual signals presents a series of substantive technical challenges, spanning data acquisition, feature representation, model architecture, and deployment considerations. These challenges, which directly motivated several of the design decisions described later in this thesis, are outlined in detail in the following subsections.

1.3.1 Variability in Facial Expressions

Facial expressions exhibit considerable inter-individual and intra-individual variability, driven by differences in age, ethnicity, gender, and culturally specific display norms governing emotional expression. Additional confounding factors, including illumination changes, head-pose variation, partial occlusion, and camera-induced artifacts, further complicate automated facial analysis. Notably, even advanced contemporary systems that integrate state-of-the-art components such as YOLOv8-based face detection together with Vision Transformer classification architectures require extensive preprocessing pipelines and aggressive data augmentation strategies in order to achieve reliable, generalizable performance [8].

1.3.2 Variability in Speech Patterns

Speech signals are characterized by substantial variability across speakers, shaped by factors such as native language, regional accent, age, and physiological differences in vocal tract anatomy. Background acoustic noise, together with the inherently ambiguous and context-dependent nature of emotional vocal expression, further complicates the task of speech emotion recognition (SER). As a direct consequence, models trained predominantly on data from specific demographic groups frequently fail to generalize effectively to broader, more heterogeneous populations, underscoring the critical need for diverse, representative, and demographically balanced training datasets [1].

1.3.3 Limited Availability of Depression-Specific Datasets

There remains a notable and persistent scarcity of publicly available datasets specifically designed and annotated for depression detection, particularly datasets that contain synchronized, high-quality audiovisual recordings paired with clinically validated depression labels. This limitation frequently necessitates reliance on proxy datasets, such as RAVDESS, in conjunction with adapted or re-mapped labeling schemes, as is the case in the present project. However, this strategy inevitably introduces a measurable domain gap between professionally acted emotional expressions and the more subtle, spontaneous, and authentic clinical manifestations of depression observed in real patient populations [9].

1.3.4 Dataset Imbalance

In binary classification settings such as the one adopted in this project, class imbalance between the “Depressed” and “Normal” categories can lead to biased models that disproportionately favor the majority class. Such bias typically manifests as inflated overall accuracy figures that mask a substantially reduced sensitivity to the clinically critical minority class. Addressing this issue effectively requires the deliberate application of targeted mitigation strategies, including systematic data augmentation techniques such as Gaussian noise injection applied at carefully calibrated intensities, and, where appropriate, class-weighted loss functions during model training [6].

1.3.5 Noise in Audio and Video Recordings

Real-world data acquisition inevitably introduces various forms of noise across both modalities under consideration. In the audio domain, ambient background noise, room reverberation, and inconsistencies in recording equipment quality can substantially degrade the integrity of extracted acoustic features. In the video domain, inconsistent lighting conditions, motion blur, and lossy compression artifacts similarly reduce the reliability and fidelity of facial feature extraction. Consequently, the design and implementation of robust preprocessing and data augmentation pipelines become an essential prerequisite for mitigating these pervasive sources of noise [9].

1.3.6 Feature-Level Fusion Challenges

Feature representations extracted independently from the audio and video modalities frequently differ substantially in terms of dimensionality, temporal resolution, and underlying statistical distribution. This heterogeneity necessitates careful normalization, temporal alignment, and dimensionality matching prior to the fusion stage. Improperly designed or insufficiently calibrated fusion strategies risk producing imbalanced joint

representations, in which one modality inadvertently dominates the combined feature space, thereby diminishing the overall effectiveness of the resulting multimodal classifier.

1.3.7 Generalization and Model Robustness

Models trained exclusively on controlled datasets composed of professionally acted emotional expressions, such as RAVDESS, may not generalize effectively when subsequently deployed in authentic, uncontrolled clinical environments. Variations in recording conditions, speaker demographic diversity, and the comparatively subtle and spontaneous nature of real-world emotional expression all present significant generalization challenges. Approaches that leverage large-scale, pre-trained transformer architectures have demonstrated considerable promise in improving cross-domain generalization under such diverse and unpredictable conditions [5].

1.3.8 Computational Complexity

The simultaneous processing of high-resolution video streams alongside corresponding audio signals imposes considerable computational demands on the underlying system, a constraint that becomes particularly acute in real-time or near-real-time application scenarios. Designing model architecture that achieves an appropriate balance between predictive accuracy and computational efficiency therefore remains an important and ongoing engineering challenge. Notably, lightweight model architectures featuring optimized parameter counts and reduced inference latency have demonstrated that high classification performance can indeed be achieved without incurring excessive computational overhead [1].

1.4 Project Objectives

1.4.1 General Objective

The primary objective of this project is to design, implement, and rigorously evaluate a multimodal deep learning system, designated MindSync, capable of automatically detecting depression-related affective indicators from synchronized facial video and speech audio signals. The proposed system is designed to classify each input instance into one of two categories, “Depressed” or “Normal,” through the use of a feature-level fusion architecture trained and validated using the RAVDESS dataset.

1.4.2 Specific Objectives

- Conduct a comprehensive and critical review of existing literature pertaining to automated depression detection, speech emotion recognition, facial expression analysis, and multimodal fusion techniques.

- Acquire, preprocess, and systematically adapt the RAVDESS dataset for binary classification purposes by re-mapping the original emotional categories and applying suitable data augmentation strategies.
- Design and implement dedicated preprocessing pipelines for both audio and video data streams, incorporating established feature-extraction techniques such as MFCC, Mel-spectrogram, Chroma features, and other relevant spectral descriptors.
- Develop dedicated deep learning-based feature-extraction modules for each modality, capable of capturing discriminative spatiotemporal patterns characteristic of depressive affect.
- Design and implement a feature-level fusion architecture that effectively integrates audio and visual feature representations into a single, unified embedding space prior to final classification.
- Train the resulting multimodal system using established techniques, including systematic data augmentation, regularization strategies, and cross-validation, in order to enhance robustness and minimize overfitting.
- Evaluate overall system performance using a comprehensive suite of standard metrics, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC), with explicit comparison against unimodal baseline models.
- Critically analyze the resulting experimental outcomes, explicitly identify the principal limitations of the proposed system, and propose well-grounded directions for future research and potential pathways toward eventual clinical deployment.

1.5 Thesis Organization

This thesis is structured into five chapters that progressively guide the reader from foundational conceptual material through detailed system development, rigorous experimentation, and critical synthesis. The organization of the content is as follows.

Chapter 1: Introduction presents the broader problem context, highlights the societal, economic, and clinical significance of Major Depressive Disorder as a global health concern, and systematically outlines the principal motivations, technical challenges, and concrete objectives underpinning the proposed MindSync system.

Chapter 2: Background and Literature Review provides a comprehensive and critically synthesized overview of existing research within the field of affective computing. This includes both traditional and deep learning-based approaches to emotion recognition, a detailed examination of established multimodal fusion strategies, a review of the datasets most commonly employed within this research area, and an exploration of explainable artificial intelligence techniques as they pertain to clinically sensitive, mental-health-oriented applications.

Chapter 3: Proposed MindSync Framework and System Design details the complete architecture and implementation of the MindSync multimodal system. It describes the end-to-end system design, the

audio processing pipeline and CNN-BiLSTM-SimpleRNN sub-network, the video processing pipeline and R3D-18-based spatiotemporal sub-network, the 568-dimensional feature-level fusion mechanism, the Multi-Layer Perceptron fusion classifier, and the rule-based Mental Health Inference Engine that maps learned emotion representations onto four clinically motivated affective indicator scores.

Chapter 4: Experimental Results and Evaluation presents a comprehensive quantitative evaluation of all three modality-specific sub-systems across two complementary experimental protocols — a primary SMOTE-augmented random split and an optimized actor-independent partition. The chapter reports per-class precision, recall, and F1-score metrics, training convergence analysis, systematic hyperparameter search outcomes, and a comparative benchmarking of MindSync results against the reference companion study and related prior work in literature.

Chapter 5: Conclusion and Future Work synthesizes the principal contributions of the thesis, distills the most substantive empirical findings and scientific insights derived from the experimental evaluation, provides an objective assessment of the methodological limitations that bound the current implementation, and presents a structured set of future development directions — including domain adaptation to clinical datasets, integration of additional physiological modalities, longitudinal deployment strategies, and the incorporation of explainability mechanisms — intended to advance the MindSync framework toward greater clinical applicability and scientific rigor.

Chapter 2

Background and Literature Review

2.1 Introduction

This chapter establishes the theoretical and empirical foundations upon which the MindSync system is constructed. It is organized into two principal sections. The first, Section 2.1, provides a critical survey of the existing literature on computational affective state recognition, examining unimodal audio-based approaches, unimodal video-based approaches, and multimodal fusion architectures, before identifying the research gap that motivates the present work. The second, Section 2.2, presents the technical background underpinning the architectural components adopted in this thesis, including three-dimensional convolutional neural networks, residual learning, and bidirectional recurrent sequence modeling. Together, these two sections situate the MindSync system within the broader research landscape and justify the specific design decisions described in the chapters that follow.

2.2 Video Modality Architecture (3D-CNN ResNet-18)

2.2.1 From Two-Dimensional to Three-Dimensional Convolution

Conventional two-dimensional Convolutional Neural Networks (2D-CNNs) operate on individual video frames treated as independent static images[13], applying learned spatial filters across the height and width dimensions of each frame in isolation. Formally, given an input feature map X of dimensions (H, W, C) , where H and W denote spatial height and width and C denotes the number of input channels, a standard 2D convolution with kernel size k produces an output feature map according to Equation.

$$Y(i, j) = \sum_p \sum_k W(p, k) X(i+p, j+k) + b \quad (2.1)$$

In this formulation, the convolution kernel W slides exclusively across the two spatial axes of a single frame, and consequently the resulting feature map Y encodes only static, frame-level spatial information. Crucially, when 2D-CNNs are applied to video data, each frame must either be processed entirely independently, discarding all temporal context, or else be artificially stacked along the channel dimension, a strategy that conflates temporal and channel information in a manner that does not respect the true temporal ordering of the sequence. This constitutes a fundamental architectural limitation for depression-indicator detection, where diagnostically relevant cues, such as delayed onset of facial expressions, reduced micro-expression frequency, and blunted

affective transitions over time, are inherently temporal phenomena that cannot be captured from any single static frame.

Three-dimensional Convolutional Neural Networks (3D-CNNs) directly address this limitation by extending the convolution operation to incorporate an explicit temporal axis. Given an input video volume X of dimensions (T, H, W, C) , where T denotes the number of frames in the temporal window, a 3D convolution with kernel size (t, k, k) is formally defined as shown in Equation.

$$Y(\tau, i, j) = \sum_r \sum_p \sum_k W(r, p, k) X(\tau+r, i+p, j+k) + b \quad (2.2)$$

Here, the kernel W simultaneously spans the temporal dimension (indexed by r), as well as the two spatial dimensions (indexed by p and k), enabling each output activation $Y(\tau, i, j)$ to integrate information from a short temporal neighborhood of consecutive frames rather than from a single isolated frame. This property allows 3D-CNNs to learn motion-sensitive filters directly from raw video volumes, capturing spatio-temporal patterns such as the rate of change of facial musculature, the velocity and acceleration of head and eyebrow movements, and the persistence or transience of micro-expressions, all of which carry diagnostic significance for affective and depressive state estimation.

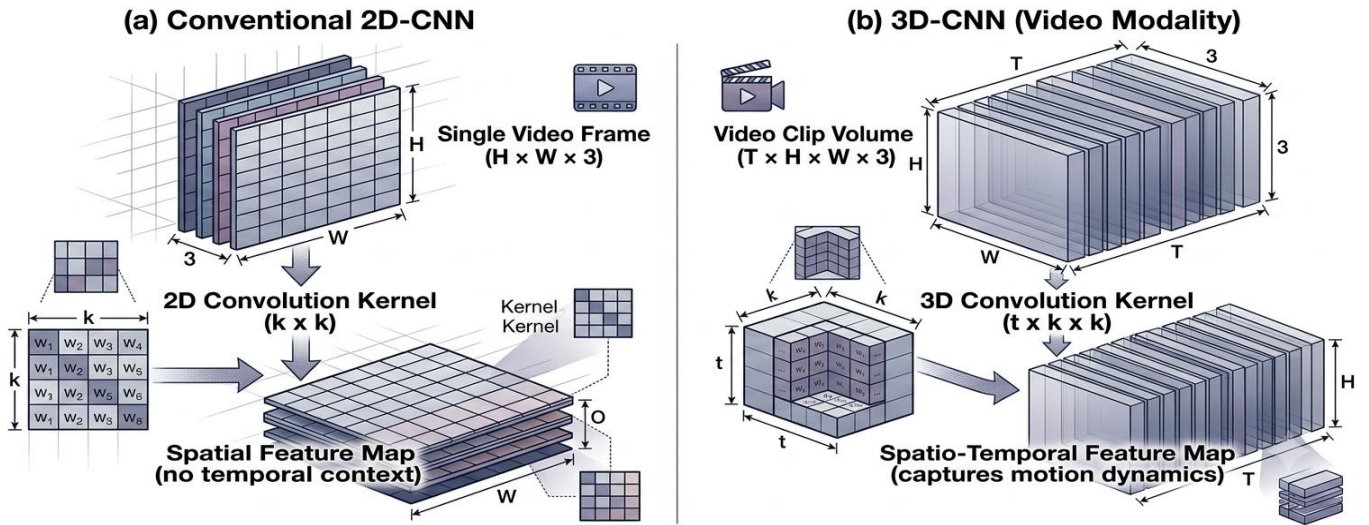


Figure 2.1: Conceptual comparison between conventional 2D-CNN frame-wise processing and 3D-CNN spatio-temporal volumetric processing.

2.2.2 Architecture of the 3D-CNN ResNet-18 (R3D-18) Backbone

While 3D convolution provides the theoretical capacity to model spatio-temporal dynamics, naively stacking deep 3D convolutional layers is well known to suffer from the vanishing-gradient problem and from

optimization difficulties that scale with network depth. The residual learning framework introduced in the ResNet family of architectures addresses this difficulty through the introduction of identity skip connections [37], which allow gradients to propagate directly across multiple layers during backpropagation. The video-modality backbone adopted in this thesis, designated R3D-18, extends the original two-dimensional ResNet-18 topology by replacing every 2D convolutional layer with its 3D counterpart, while preserving the residual block structure that characterizes the ResNet family [38].

Formally, each residual block computes an output $H(x)$ as the elementwise sum of an identity mapping and a learned residual function $F(x)$, as expressed in Equation.

$$H(x) = F(x) + x \quad (2.3)$$

where $F(x)$ denotes the transformation learned by a small stack of 3D convolutional layers, each followed by 3D batch normalization and a Rectified Linear Unit (ReLU) non-linearity. This residual formulation ensures that, in the worst case, a given block can learn to approximate the identity function, $F(x) \rightarrow 0$, thereby guaranteeing that the addition of further depth does not degrade representational capacity relative to a shallower network, a property that is particularly valuable when training on comparatively modest datasets such as the re-labeled RAVDESS corpus employed in this thesis.

The complete R3D-18 pipeline adopted for the video modality proceeds as follows. The raw input to the network consists of a fixed-length clip of sixteen consecutively sampled frames, each resized to a spatial resolution of 112 by 112 pixels across three RGB channels, yielding an input volume of dimensions $16 \times 112 \times 112 \times 3$. This volume is first processed by a 3D stem convolution employing a $7 \times 7 \times 7$ kernel with stride 2, which performs an initial, aggressive spatio-temporal down sampling while extracting low-level edge- and motion-sensitive features. The resulting feature volume is subsequently propagated through four sequential stages of residual blocks, with channel depths of 64, 128, 256, and 512 respectively, each stage progressively halving the spatial and temporal resolution of the feature maps while correspondingly doubling the channel depth, in accordance with the standard ResNet design philosophy of trading spatial resolution for representational richness as network depth increases.

Following the final residual stage, a 3D Global Average Pooling (GAP) operation collapses the remaining spatial and temporal dimensions into a single value per channel, yielding a compact, fixed-length 512-dimensional

feature vector that constitutes the final video-modality representation. This pooling operation is formally defined in Equation.

$$v^c = (1 / (T' \times H' \times W')) \sum_t \sum_i \sum_j F^c(t, i, j) \quad (2.4)$$

where F^c denotes the activation of the final feature map at channel c , and T' , H' , and W' denote the reduced temporal and spatial dimensions remaining after the four residual stages. The resulting 512-dimensional vector v constitutes a compact yet highly discriminative spatio-temporal embedding of the input video clip, encoding both the static appearance of the subject's face and the dynamic evolution of facial musculature across the observed temporal window. This embedding is subsequently passed through a final fully connected layer that produces an 8-dimensional SoftMax probability distribution over emotion categories, while the 512-dimensional feature vector itself is additionally retained and forwarded, unmodified, to the multimodal feature-fusion stage described in later sections of this thesis.

It is important to emphasize that the diagnostic value of this architecture for depression-indicator detection lies precisely in its ability to jointly model spatial appearance and temporal dynamics within a single, unified representation. Depressive effect is frequently characterized not by any single static facial configuration, but by characteristic temporal patterns, such as a reduced rate of spontaneous smiling onset, prolonged dwell time in neutral or low-arousal facial configurations and diminished overall facial expressivity across the duration of an interaction. The 512-dimensional embedding produced by the R3D-18 backbone is explicitly designed to encode precisely this class of temporally extended behavioral signature, distinguishing it fundamentally from purely frame-based, static facial-expression classifiers.

2.3 Audio Modality Architecture (MFCC + CNN + BiLSTM + MLP)

2.3.1 Audio Preprocessing and MFCC Feature Extraction

The acoustic processing pipeline adopted for the audio modality of the MindSync system begins with a series of well-established preprocessing operations designed to transform the raw speech waveform into a compact, perceptually meaningful feature representation. The raw waveform is first segmented into overlapping short-duration frames, typically spanning twenty to twenty-five milliseconds with a hop size of approximately ten milliseconds, an interval chosen to satisfy the quasi-stationarity assumption that underlies short-time spectral analysis of speech, given that the physical configuration of the vocal tract changes comparatively slowly

relative to this timescale. A pre-emphasis filter is subsequently applied to amplify high-frequency spectral components that are otherwise attenuated by the natural spectral roll-off of voiced speech, thereby improving the discriminability of the resulting spectral features.

Each resulting frame is next windowed using a Hamming function to mitigate spectral leakage artifacts that would otherwise arise from the implicit rectangular windowing of a finite-length signal segment and is then transformed into the frequency domain via the Short-Time Fourier Transform (STFT), as expressed in Equation.

$$X(m, k) = \sum_n x(n) w(n) e^{-j2\pi kn/N} \quad (2.5)$$

where $x(n)$ denotes the input waveform, $w(n)$ denotes the Hamming window function, m indexes the analysis frame, k indexes the discrete frequency bin, and N denotes the frame length in samples. The resulting magnitude spectrum is subsequently passed through a bank of triangular Mel-scaled filters, designed to approximate the non-linear frequency resolution of human auditory perception, in which frequency discrimination is considerably finer at low frequencies than at high frequencies. The Mel-filtered energy within each filter band m is computed as shown in Equation.

$$S(m) = \sum_k |X(m, k)|^2 H_k \quad (2.6)$$

where H_k denotes the triangular weighting function associated with the k -th Mel filter. Finally, the Mel-Frequency Cepstral Coefficients (MFCCs) themselves are obtained by applying a Discrete Cosine Transform (DCT) to the logarithm of these Mel-filtered energies [34], a step that serves both to compress the dynamic range of the resulting features, in a manner consistent with the logarithmic sensitivity of human loudness perception, and to decorrelate adjacent filter-bank energies, thereby producing a compact and statistically efficient feature representation. This final transformation is given in Equation.

$$c_n = \sum_m \log(S(m)) \cos [n(m - 0.5)\pi/M] \quad (2.7)$$

where M denotes the total number of Mel filter banks and indexes the resulting cepstral coefficient. In the implementation adopted for this thesis, the first forty cepstral coefficients are retained per analysis frame, yielding a compact forty-dimensional MFCC feature vector that serves as the principal acoustic representation forwarded to the subsequent deep learning stages of the audio pipeline, and which is additionally retained, unmodified, for direct inclusion within the multimodal feature-fusion vector described in later chapters.

2.3.2 Convolutional Feature Extraction over MFCC Sequences

Once the sequence of frame-level MFCC vectors has been computed, the resulting time series is propagated through a stack of one-dimensional Convolutional Neural Network (CNN) layers, whose function within this pipeline is to learn local, translation-invariant spectral patterns that recur across the temporal extent of the speech signal, such as characteristic formant transitions, abrupt spectral discontinuities associated with consonant articulation, and short-duration prosodic inflections. Formally, a one-dimensional convolution operating over the MFCC sequence is defined in Equation.

$$h_i = \sum_p w_p \cdot x_{i+p} + b \quad (2.8)$$

where x denotes the input MFCC sequence, w_p denotes the learnable convolution kernel weights, and h_i denotes the resulting activation at temporal position i . Each convolutional layer is followed by a Rectified Linear Unit (ReLU) non-linearity, a max-pooling operation that progressively reduces the temporal resolution of the feature sequence while retaining the most salient local activations, and a batch normalization step that stabilizes training by normalizing the distribution of layer activations. Successive convolutional blocks employ progressively larger filter counts, typically increasing from 64 to 512 filters across four stacked blocks, thereby allowing the network to learn an increasingly abstract and hierarchical representation of the underlying acoustic signal, in a manner directly analogous to the hierarchical spatial feature learning performed by the video-modality 3D-CNN backbone described in Section 2.2.

2.3.3 Bidirectional LSTM for Temporal and Contextual Modeling

While the convolutional layers described above are well suited to extracting local, short-range spectral patterns, depressive speech is additionally characterized by longer-range prosodic phenomena, including reduced variability in fundamental frequency contours across an entire utterance, systematically slower speaking rates, and an increased frequency and duration of inter-word pauses, phenomena that necessarily span temporal windows considerably longer than the receptive field of any individual convolutional filter. To capture such extended temporal dependencies, the convolutional feature sequence is subsequently propagated through Bidirectional Long Short-Term Memory (BiLSTM) layers.

The standard LSTM cell update equations governing the forward-direction hidden state at time step t are given by Equation.

$$h_t = \sigma(W_h h_{t-1} + W_x x_t + b_h) \quad (2.9)$$

where x_t denotes the input feature vector at time step t , h_{t-1} denotes the hidden state retained from the preceding time step, and W_h , W_x , and b_h denote the trainable recurrent weight matrices and bias term governing the cell's internal gating mechanisms, which jointly regulate the retention, update, and forgetting of information across the temporal sequence [36]. Crucially, the bidirectional formulation employed in this thesis computes two such hidden-state sequences in parallel: a forward pass that processes the MFCC-derived feature sequence in its natural chronological order, and a backward pass that processes the identical sequence in reverse temporal order. The final BiLSTM output at each time step is obtained by concatenating the corresponding forward and backward hidden states, as expressed in Equation.

$$h_t^{bi} = [h_t \rightarrow; h_t \leftarrow] \quad (2.10)$$

This bidirectional architecture is of particular importance for depression-related speech analysis, since prosodic abnormalities such as a flattened pitch contour or an unusually long pause are frequently only interpretable in light of both their preceding and their subsequent acoustic context; a momentary pause, for instance, acquires depressive significance only when considered in relation to the surrounding speech rate and intonation pattern. By integrating information from both temporal directions simultaneously, the BiLSTM layers are able to construct a substantially richer and more context-sensitive representation of the speech signal than either a unidirectional LSTM or a purely convolutional architecture could achieve in isolation. The complete audio-modality pipeline, encompassing MFCC extraction, convolutional feature learning, and bidirectional temporal modeling, collectively constitutes a computationally efficient yet expressively powerful sequence of transformations that progressively abstracts the raw speech signal into a rich, depression-discriminative representation, directly preparing the learned embedding for integration within the multimodal feature-fusion stage described in Chapter 3.

2.3.4 The Role of the Multi-Layer Perceptron (MLP) Classifier

Following the bidirectional temporal modeling stage, the resulting sequence of BiLSTM hidden states is reduced to a single, fixed-length representation, typically through the use of the final hidden state of the sequence or through an aggregation operation such as temporal pooling and is subsequently passed through one or more fully connected Multi-Layer Perceptron (MLP) layers. The role of this MLP component within the audio pipeline is to perform a final, non-linear transformation of the learned temporal representation into a form suitable for classification, effectively learning a decision boundary within the high-dimensional feature space constructed by the preceding convolutional and recurrent layers.

Each MLP layer computes its output according to the standard fully connected transformation given in Equation.

$$z = \varphi (W z' + b) \quad (2.11)$$

where z' denotes the input activation vector received from the preceding layer, W and b denote the learnable weight matrix and bias vector of the fully connected layer, and ϕ denotes a non-linear activation function, typically the Rectified Linear Unit (ReLU) for intermediate layers. Within the audio-only branch of the system, the final MLP layer culminates in an 8-dimensional SoftMax output, producing a normalized probability distribution over the eight emotion categories present in the RAVDESS taxonomy. Critically, in addition to this unimodal classification output, both the 128-dimensional fully connected audio embedding and the resulting 8-dimensional SoftMax probability vector are retained and forwarded, unmodified, to the multimodal feature-fusion stage, where they are concatenated with their video-modality counterparts to form the unified 568-dimensional joint representation that underlies the final depression-indicator classification performed by the complete MindSync system, as will be described in full architectural detail in the chapters that follow. The complete audio-modality

Taken together, the audio pipeline described in this section, comprising MFCC-based spectral feature extraction, hierarchical convolutional pattern learning, bidirectional temporal context modeling, and a final fully connected classification stage, constitutes a computationally efficient yet expressive alternative to large-scale self-supervised transformer architectures, directly addressing the deploy ability concerns identified in the literature review of Section 2.1, while remaining fully compatible with, and complementary to, the spatio-temporal video-modality representation produced by the 3D-CNN ResNet-18 backbone described in Section 2.2.

2.4 Literature Review of mental health

Paul, Saha, and Hossain (2025) proposed EmoAugNet, a hybrid one-dimensional convolutional neural network and Long Short-Term Memory (CNN-LSTM) architecture designed specifically to address the data scarcity and limited generalization that commonly affect speech emotion recognition systems. The methodology centers on a signal-level augmentation pipeline that systematically combines noise injection, pitch shifting, and time stretching in a novel combinatorial manner, rather than applying these transformations in isolation as in prior work. Acoustic descriptors, namely Root Mean Square Energy (RMSE), Mel-Frequency Cepstral Coefficients (MFCC), and Zero-Crossing Rate (ZCR), are extracted from both the original and augmented waveforms before being passed through stacked one-dimensional convolutional layers for local spectral pattern extraction,

followed by LSTM layers responsible for capturing the longer-range temporal dependencies inherent to emotional prosody. The authors evaluated their framework on the IEMOCAP and RAVDESS benchmark datasets, reporting weighted accuracies ranging from approximately 94.5% to 96.8%, figures that exceeded several previously published CNN-LSTM baselines and substantiated the central claim that combinatorial signal-level augmentation meaningfully improves model robustness against acoustic variability.

Chowdhury, Banik, Hoque, and Banerjee (2025-2026) introduced a hybrid Deep Conditional Random Field-Bidirectional LSTM (DCRF-BiLSTM) framework that integrates engineered acoustic descriptors, including MFCCs, prosodic features, and spectral features, with a sequence-modeling architecture explicitly designed to capture both local emotional cues and their broader contextual dependencies. The conditional random field component is incorporated to regularize the temporal labeling process, encouraging the model to learn smoother and more coherent transitions across emotional states throughout an utterance, while the bidirectional LSTM layers preserve information flow in both forward and backward temporal directions. A particularly notable aspect of the study's methodology is its breadth of evaluation: the authors tested their model not only on the RAVDESS corpus in isolation, where it achieved an accuracy of 97.83%, but also across multiple combined cross-corpus configurations spanning up to five distinct datasets simultaneously, obtaining a combined accuracy of 93.76%. This cross-corpus evaluation strategy is comparatively rare within the existing literature and provides meaningful evidence regarding the model's capacity, and corresponding limitations, when generalizing beyond a single recording environment or speaker population.

Vasuki, Padhma Priya, and Pooja (2024) examined the problem of emotion recognition from a multimodal perspective, proposing a fusion architecture that jointly leverages audio and video signals rather than relying on either modality in isolation. Their methodology combines an audio-processing branch built upon a CNN-RNN-BiLSTM pipeline operating over MFCC features with a video-processing branch employing a three-dimensional convolutional network based on the ResNet-18 topology to extract spatio-temporal facial representations. The outputs of these two modality-specific branches are subsequently merged through a joint multi-layer perceptron (MLP) classifier, allowing the model to exploit complementary information present across both channels. The authors reported that their multimodal fusion approach consistently outperformed comparable unimodal baselines across cross-validation folds, reinforcing the broader consensus within the field that audio-visual fusion strategies yield more robust emotion-recognition outcomes than single-modality systems, although the publicly available abstract of the study did not specify a single consolidated accuracy figure.

Gavali and Verma (2025) investigated the application of self-supervised transformer architectures to speech emotion recognition, focusing specifically on the potential benefits of ensemble-based decision aggregation. Their methodology involved benchmarking several large pre-trained self-supervised models, most notably variants of Wav2Vec2.0-XLSR-53, individually on the RAVDESS dataset before combining their predictions through a majority-voting mechanism across both three-model and five-model ensemble configurations. The best-performing individual model achieved an accuracy of 93%, while the five-model ensemble achieved a substantially higher accuracy of 96.88%, demonstrating that aggregating the predictions of multiple self-supervised backbones can yield meaningful performance gains relative to any single constituent model. However, the authors also noted that this performance improvement is accompanied by a considerable increase in computational and memory overhead, given that each transformer backbone within the ensemble independently comprises tens of millions of parameters, a trade-off with direct implications for real-time or resource-constrained deployment scenarios.

Waleed and Shaker (2025) proposed a lightweight, multi-branch one-dimensional convolutional neural network explicitly designed to balance recognition accuracy against computational efficiency. The architecture processes three complementary acoustic feature streams, namely MFCCs, Mel-spectrograms, and Chroma vectors, through separate parallel convolutional branches before merging their learned representations at the feature level through a dedicated fusion layer. This design choice allows the network to exploit the distinct but complementary information captured by each feature type without requiring the substantially larger parameter counts associated with transformer-based architectures. Evaluated on the RAVDESS and MELD datasets, the proposed model achieved accuracy of 91.9% and 94.0%, respectively, while maintaining a comparatively small footprint of approximately 1.2 million parameters and 115 million floating-point operations, figures considerably lower than those typically reported for self-supervised transformer ensembles. The authors positioned their approach as particularly well-suited to resource-constrained or embedded deployment contexts, while acknowledging that the audio-only design remains susceptible to misclassification in cases involving nuanced or blended emotional expressions.

Sareen and Seeja (2025) approached emotion recognition from a purely visual standpoint, proposing a pipeline that integrates a YOLOv8-based face detector with a Vision Transformer (ViT) classifier for facial emotion recognition from video data. The methodology begins with the extraction of a representative middle frame from each video clip, from which the YOLOv8 detector localizes the facial region prior to classification. The cropped

facial region is then passed to a pre-trained Vision Transformer, fine-tuned to classify the input into one of several discrete emotion categories. To improve robustness against variations in lighting and orientation, the authors applied a series of data augmentation techniques, including horizontal flipping, brightness adjustment, and the injection of Gaussian noise, during the model training process. Evaluated on the RAVDESS dataset, the proposed YOLOv8-ViT pipeline achieved an accuracy of 90.81%, a result the authors reported as surpassing several previously published facial emotion recognition systems, while noting that the reliance on a single extracted frame per clip may limit the model's sensitivity to temporally extended emotional dynamics such as gradual expression onset or offset.

The reviewed studies reveal a clear empirical trajectory from single modality toward multimodal architectures, yet the majority remain constrained by their exclusive reliance on acoustic signals. Chowdhury et al. (2025–2026) achieve the highest reported single-corpus accuracy of 97.83% on RAVDESS and 93.76% across five combined datasets, while Gavali and Verma (2025) push audio-only performance to 96.88% through self-supervised transformer ensembles. Despite these impressive figures, both systems are fundamentally limited to the acoustic domain, rendering them incapable of capturing the visual behavioral cues — micro-expressions, facial dynamics, and postural indicators — that carry particular diagnostic salience for depression detection. Among all surveyed contributions, Vasuki, Padhma Priya, and Pooja (2024) present the most architecturally sound approach, not by reporting the highest accuracy, but by being the only system to integrate audio and visual modalities within a unified end-to-end pipeline, directly overcoming the unimodal information ceiling that constrains every other reviewed method. This synthesis establishes that raw accuracy on a single corpus and model scale are insufficient criteria for architectural superiority in the context of affective depression screening; what is required is the joint exploitation of heterogeneous behavioral signals — precisely the design principle upon which the MindSync framework is founded. **Table 2.1** below presents a structured comparative summary of all reviewed studies across the dimensions of methodology, model architecture, feature representation, empirical results, and principal limitations.

Table 2.1 Comparison table of Literature Review

Author & Year	Methodology	Model	Features	Results	Limitations	Link
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Waleed & Shaker2025 · MDPI Information	Multi-branch feature-level fusion; parallel 1D-CNN streams per feature type; data augmentation (Gaussian noise + time shifting)	Lightweight 1D-CNN with 3 parallel branches, fully connected layers, SoftMax output (~1.2M params, 115 MFLOPs)	- MFCC (40 coefficients) - Mel-spectrogram (128 bins) Chroma vectors (12 dims)	RAVDESS: 91.9% MELD: 94.0% F1: 90.8% / 93.5%	Audio-only; vulnerable to nuanced/blended emotions; not tested on noisy real-world data or cross-corpus scenarios	Link
Paul, Saha & Hossain2025 · arXiv:2508.06321	Hybrid 1D-CNN + LSTM with signal-level augmentation pipeline (noise addition, pitch shifting, time stretching + novel combinatorial augmentation)	EmoAugNet : 1D-CNN layers followed by LSTM layers; ReLU and ELU activation variants	- RMSE (root mean square energy) - MFCC ZCR (zero-crossing rate)	IEMOCAP WA: 94.53–96.75% RAVDESS WA: 93.72–95.78% Unweighted: 91.28–94.98%	Augmentation pipeline adds preprocessing overhead; audio-only scope; combinatorial augmentation complexity may limit reproducibility	Link
Chowdhury, Banik, Hoque & Banerjee2025–2026 · arXiv:2507.07046	Hybrid DCRF-BiLSTM with engineered feature set; evaluated across 5 individual datasets and multiple combined dataset configurations	DCRF-BiLSTM : Dilated Convolutional Recurrent Filter combined with Bidirectional LSTM	- MFCC - Prosodic features Spectral features	RAVDESS: 97.83% TESS / EMO-DB: 100% 5-dataset combo: 93.76% R+T+S combo: 98.82%	Performance degrades on cross-corpus combination; audio-only; generalization across unseen domains not established	Link
Gavali & Verma2025 · IJDDMM Vol.17 No.2	Majority-voting ensemble of large self-supervised transformers; 3-model and 5-model ensemble configurations on RAVDESS	SSL transformer ensemble (Wav2Vec2.0-XLSR-53 as best single model; majority voting across 6 candidate models)	Self-supervised audio embeddings (raw waveform → learned representations)	Single best model: 93% 5-model ensemble: 96.88% 3-model ensemble: 96.53%	Audio-only; very high computational and memory footprint (large transformer models); limited scalability to real-time or edge deployment	Link

Vasuki, Padhma Priya & Pooja2025 · SSRN:5274911	Multimodal audio-video fusion; audio branch via CNN-RNN-BiLSTM; video branch via 3D CNN (ResNet-18); joint MLP classifier	Multimodal fusion: audio CNN-RNN-BiLSTM + video 3D-ResNet-18 → MLP joint classifier	Audio: MFCC features Video: spatial-temporal features from 3D CNN (ResNet-18)	Results reported on cross-validation folds Outperforms unimodal baselines; no single accuracy figure published in abstract	Requires synchronized audio-video data; higher architectural complexity; no specific accuracy benchmark cited in abstract; dataset scope not clearly specified	Link
EPJ / IGDTUW authors2025 · EPJ Web Conf. 328, 01040	Video-based facial emotion recognition; YOLOv8 for face detection on middle frame extraction; ViT for emotion classification; data augmentation (flipping, brightness, Gaussian noise)	YOLOv8 + ViT (Vision Transformer): face detection → pre-trained ViT for 7-class emotion classification	Spatial-temporal facial features from video frames Middle frame extraction from RAVDESS video clips	RAVDESS: 90.81% Surpasses several existing FER state-of-the-art systems	Video/facial modality only — no audio; single-frame (middle frame) extraction may miss temporal emotion dynamics; domain limited to acted facial expressions in RAVDESS	Link

As is evident from Table 2.1, the overwhelming majority of existing contributions address either the audio or the video modality in isolation, or else employ multimodal fusion strategies whose computational complexity, owing to large transformer backbones or adaptively restructured network topologies, limits their practical deployability. Moreover, none of the surveyed studies directly target depression-indicator detection through a binary re-mapping of the RAVDESS emotional taxonomy; rather, they remain focused on multi-class emotion recognition as an end in itself. The MindSync system proposed in this thesis is explicitly positioned to address this gap by combining a computationally efficient 3D-CNN ResNet-18 video backbone with a hybrid CNN-BiLSTM audio pipeline within a unified feature-level fusion architecture, explicitly re-purposed and re-labeled for the clinically motivated task of automated depression-indicator screening.

Chapter 3

Proposed MindSync Framework and System Design

3.1 Introduction

Building directly upon the conceptual motivation established in Chapter 1 and the theoretical and architectural foundations surveyed in Chapter 2, this chapter presents the complete engineering design and implementation of the proposed multimodal system, designated MindSync. Whereas the preceding chapters were primarily concerned with situating the present work within the broader scientific literature and establishing the mathematical principles underlying three-dimensional convolutional video modeling and hybrid CNN-BiLSTM audio modeling, this chapter is concerned with the concrete, end-to-end realization of these principles as a functioning computational pipeline. The chapter proceeds from the dataset acquisition and exploratory analysis stage, through the design of the modality-specific audio and video sub-networks, to the construction of the joint feature-level fusion representation and the corresponding fusion classifier and concludes with a description of the training configuration and software environment employed throughout the implementation.

3.2 Overview of the Proposed MindSync Framework

The MindSync framework is conceived as a modular, three-stage multimodal pipeline in which two independent modality-specific deep learning sub-systems, an audio branch and a video branch, are first trained and evaluated in isolation, before their learned internal representations are subsequently combined through a dedicated feature-level fusion mechanism that feeds a final fusion classifier. This staged design philosophy offers two principal engineering advantages. First, it allows each modality-specific sub-network to be optimized, validated, and diagnosed independently, considerably simplifying debugging and hyperparameter tuning relative to monolithic, end-to-end multimodal architecture trained from scratch. Second, it permits the resulting modality-specific embeddings, namely the 40-dimensional MFCC descriptor together with its associated 8-dimensional audio Softmax distribution, and the 512-dimensional spatio-temporal video embedding together with its associated 8-dimensional video Softmax distribution, to be reused, inspected, and benchmarked independently of the downstream fusion stage, a property that proved valuable during the iterative development of the system.

The complete pipeline operates on the RAVDESS corpus, paired such that every sample is associated with both a speech or song audio recording and its corresponding synchronized video recording of the performing actor. Figure 3.1 below presents a system-level overview of the proposed MindSync pipeline, illustrating the flow of

information from the raw paired audio-visual input, through the two independent modality-specific branches, to the construction of the unified 568-dimensional fusion vector and the final classification output. A detailed technical exposition of each individual stage depicted in the figure is provided in the sections that follow.

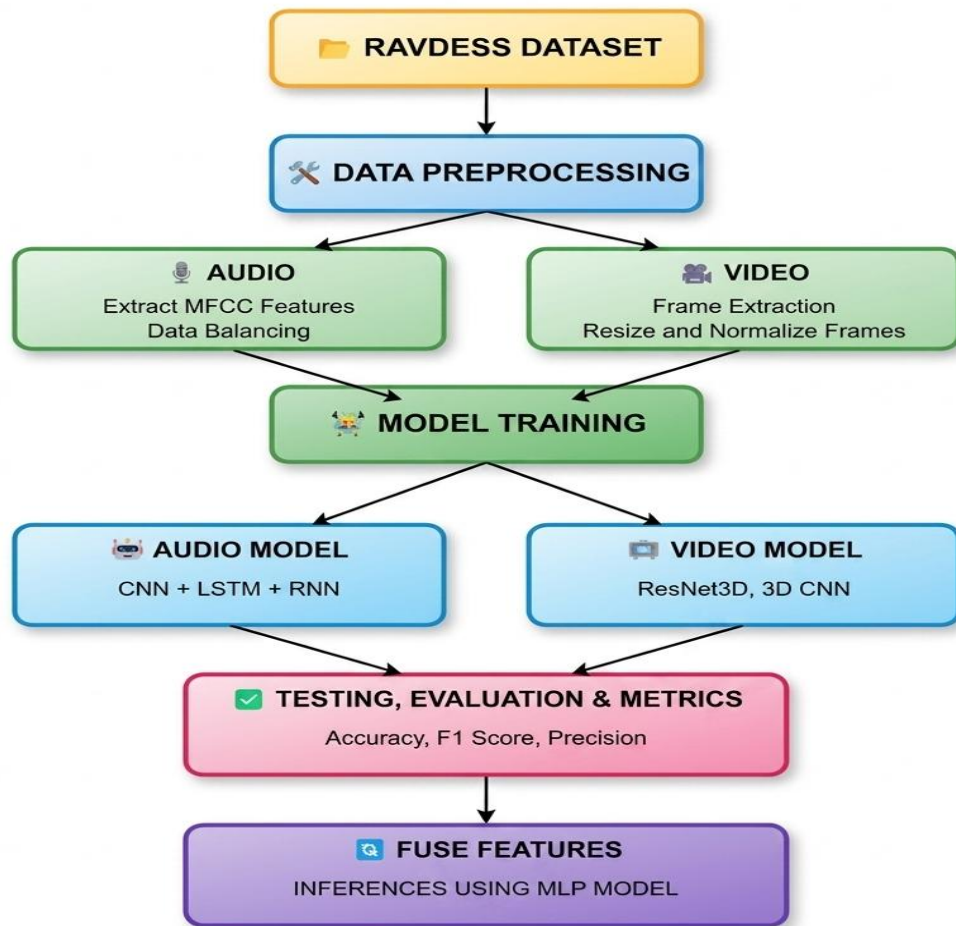


Figure 3.1: System-level overview of the proposed MindSync end-to-end multimodal pipeline, from paired audio-visual input to fusion-based classification.

3.3 Dataset Acquisition and Manifest Construction

The implementation begins with the programmatic acquisition of the RAVDESS corpus through the KaggleHub distribution channel, followed by the construction of a unified, paired data manifest. Because the RAVDESS corpus separately distributes its speech and song recordings, and because each utterance is independently associated with both an audio-only file and a corresponding full audio-visual file, the manifest-construction stage is responsible for systematically traversing the extracted directory structure, parsing the

structured RAVDESS filename convention described in Chapter 2, and assembling a single tabular manifest in which every row corresponds to one fully paired audio-video sample, annotated with its associated emotion label, actor identifier, and inferred gender, the latter being deterministically derivable from the parity of the actor index. This manifest construction step, implemented in the `build_manifest` routine of the underlying data module, is foundational to the remainder of the pipeline, since every subsequent preprocessing, training, and fusion operation is indexed directly against the resulting paired manifest, ensuring strict sample-level correspondence between the audio and video modalities at every stage of the system.

Both the speech and song subsets of RAVDESS are retained and combined within the manifest, in contrast to several prior studies surveyed in Chapter 2 that restrict their analysis to the speech subset alone. This design decision was adopted in order to— maximize the total number of available paired training samples and to expose the modality-specific sub-networks to a broader range of vocal and facial expressivity, encompassing both spoken and sung emotional delivery, which is hypothesized to improve the generalization capacity of the resulting feature representations.

3.4 Exploratory Data Analysis of the Paired Manifest

Prior to model development, a structured exploratory data analysis (EDA) stage is performed on the constructed manifest, serving both to verify the correctness of the preceding data-acquisition pipeline and to characterize the underlying statistical properties of the dataset that subsequently inform downstream design decisions, most notably the decision to apply class-balancing techniques during audio model training. This analysis examines the categorical distribution of emotion labels across the eight RAVDESS emotion classes, the distribution of samples across the male and female actor sub-populations, and the overall structure of the paired audio-video manifest. The resulting distributional characteristics confirm that, while the RAVDESS corpus is reasonably well balanced across its eight emotion categories by experimental design, residual class imbalance nonetheless persists following the manifest-pairing and validation-filtering operations, directly motivating the Synthetic Minority Over-sampling Technique (SMOTE) strategy described in Section 3.6.

3.5 Audio Pipeline: Preprocessing, Feature Extraction, and Model Architecture

3.5.1 Audio Preprocessing and MFCC Feature Extraction

Consistent with the theoretical exposition presented in Section 2.3, the audio preprocessing stage of the implemented system begins with the loading of each raw waveform at a fixed sampling rate, followed by the application of a pre-emphasis filter that amplifies high-frequency spectral content according to the standard

first-order difference relation $y'[n] = y[n] - 0.97 \times y[n-1]$, where the coefficient 0.97 was selected in accordance with established practice in speech-processing literature. The pre-emphasized waveform is subsequently transformed into a forty-dimensional Mel-Frequency Cepstral Coefficient (MFCC) representation through a short-time Fourier analysis employing a Hamming window, a 512-point Fast Fourier Transform, and a hop length of 160 samples, directly implementing Equations (2.5) through (2.7) introduced in Chapter 2. The resulting frame-level MFCC matrix is subsequently aggregated into a single fixed-length forty-dimensional feature vector per audio sample, which constitutes the principal input representation supplied to the audio classification sub-network described in the following subsection.

To facilitate qualitative inspection of this preprocessing pipeline during development, the implementation additionally generates diagnostic visualizations comparing the raw waveform, the pre-emphasized waveform, and the resulting MFCC spectrogram for representative samples drawn from each emotion category, a practice that proved valuable in confirming the absence of preprocessing artifacts prior to full-scale feature extraction across the complete dataset.

3.5.2 Audio Model Architecture: CNN, BiLSTM, and SimpleRNN

Following extraction, the forty-dimensional MFCC vectors are reshaped into a sequence-compatible tensor format and supplied to the audio classification sub-network, implemented through the `build_audio_model` constructor of the underlying model module. The network begins with four sequential one-dimensional convolutional blocks employing 64, 128, 256, and 512 filters respectively, each block incorporating max-pooling and batch normalization in accordance with the design rationale presented in Section 2.3.2.

The resulting convolutional feature sequence is subsequently propagated through two stacked Bidirectional LSTM layers, comprising 256 and 512 units respectively. The bidirectional formulation was deliberately preferred over a unidirectional LSTM architecture because depressive prosodic abnormalities — such as unusual pause durations, flattened pitch contours, and reduced speech rate — are most reliably identified in light of both prior and subsequent acoustic context simultaneously; a momentary pause or intonation change, for instance, acquires its diagnostic significance only when interpreted relative to the surrounding speech pattern in both temporal directions, a property that a forward-only recurrent pass fundamentally cannot provide.

Following the BiLSTM layers, a Bidirectional SimpleRNN layer of 512 units provides an additional stage of lightweight sequential refinement at reduced computational cost, capturing residual short-range temporal dependencies not fully resolved by the deeper LSTM gating mechanisms. The recurrent output is finally reduced

through two fully connected dense layers of 128 and 64 units, culminating in an eight-way Softmax classification layer corresponding to the eight RAVDESS emotion categories. Figure 3.2 below illustrates the complete sequential structure of this audio sub-network.

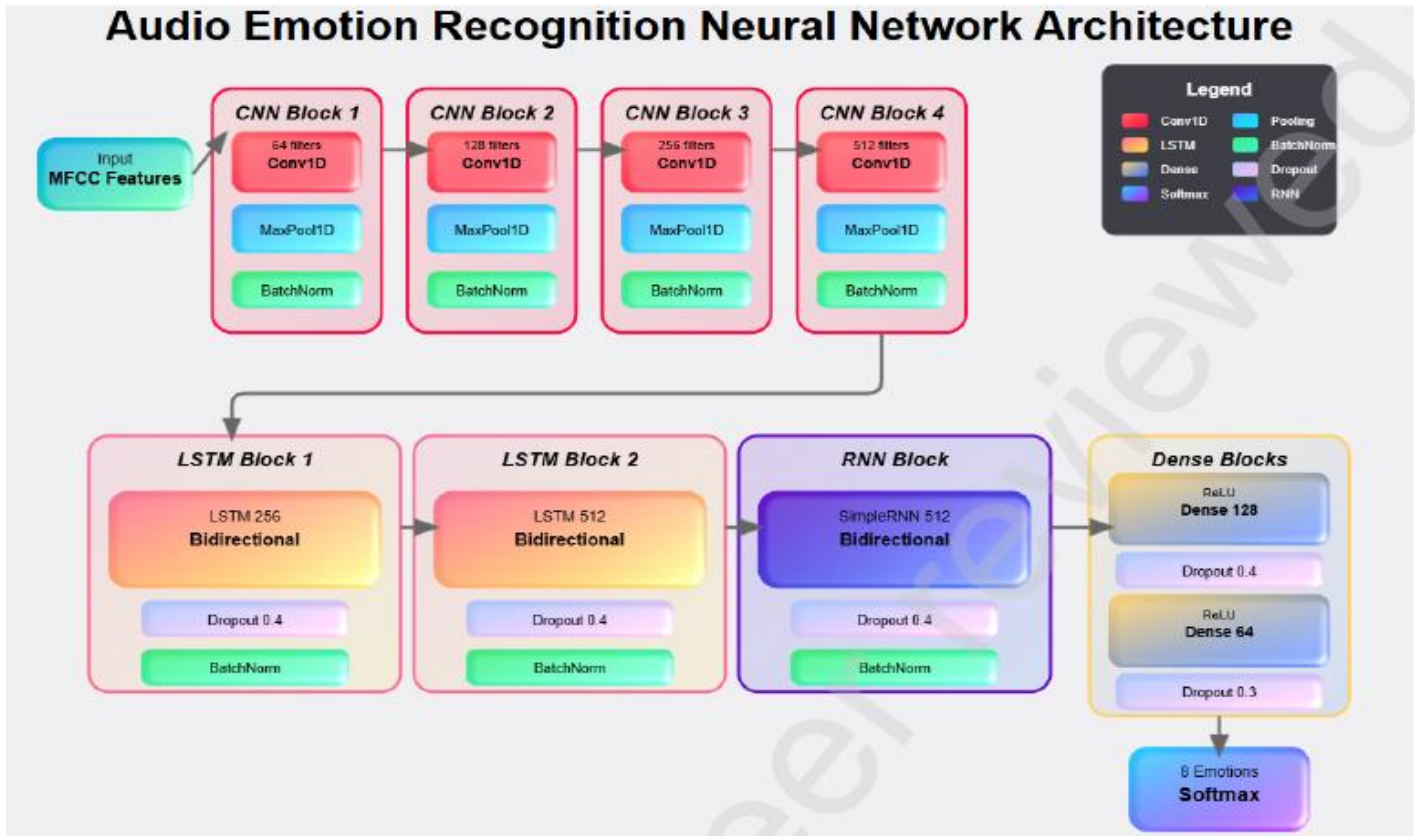


Figure 3.2: Implemented CNN + BiLSTM + SimpleRNN architecture of the MindSync audio-modality emotion-recognition sub-network.

The audio sub-network is compiled using the Adam optimizer together with the sparse categorical cross-entropy loss function, and is trained subject to a ReduceLROnPlateau learning-rate scheduler that monitors validation loss, an EarlyStopping callback that monitors validation accuracy and restores the best-performing weights upon termination, and a ModelCheckpoint callback that persists the best-performing model parameters to disk throughout training, collectively implementing the regularization and optimization strategy detailed in Section 2.3 and Table 1.IV.

3.6 Video Pipeline: Preprocessing and Pretrained R3D-18 Architecture

3.6.1 Video Preprocessing and Clip Construction

Each video sample associated with the paired manifest is processed through a dedicated preprocessing routine that uniformly samples a fixed-length sequence of sixteen frames from the source recording, applies

facial localization and cropping to suppress irrelevant background content, resizes the resulting frames to a spatial resolution of 112 by 112 pixels, and normalizes pixel intensities prior to tensor conversion, directly implementing the video-preprocessing procedure formalized in Equation (4) of the underlying methodological framework and Section 2.2 of Chapter 2. The resulting tensor, of shape corresponding to three color channels, sixteen temporal frames, and a 112-by-112 spatial resolution, constitutes the direct input to the video classification sub-network. Samples for which facial localization fails, owing to occlusion, extreme pose, or corrupted source video, are substituted with a zero-valued tensor of identical shape within the custom PyTorch Dataset implementation, ensuring that batch construction remains well defined throughout training even in the presence of occasional preprocessing failures.

3.6.2 Video Model Architecture: Pretrained R3D-18 with Auxiliary 3D Convolutional Refinement

In a deliberate refinement of the architectural template introduced in Section 2.2, the video sub-network implemented within the MindSync framework employs an R3D-18 backbone that has been pretrained on the large-scale Kinetics-400 action-recognition corpus, rather than a backbone trained from random initialization. This transfer-learning strategy was adopted in direct response to the comparatively limited scale of the RAVDESS video subset relative to the substantial parameter count of the R3D-18 architecture, and is consistent with well-established practice in the video-understanding literature, wherein large-scale action-recognition pretraining has been repeatedly shown to provide highly transferable low-level and mid-level spatio-temporal feature representations applicable to downstream tasks, including facial affect recognition, for which large labeled video corpora are considerably scarcer than for generic action recognition.

The pretrained R3D-18 backbone produces a 512-dimensional spatio-temporal embedding for each input clip, identical in dimensionality to the representation described theoretically in Section 2.2.2. This embedding is subsequently refined through three additional, task-specific three-dimensional convolutional layers employing 256, 128, and 64 filters respectively, allowing the network to adapt the generic, action-recognition-oriented Kinetics-400 representation to the considerably more fine-grained and subtle spatio-temporal patterns characteristic of facial emotional expression. The refined feature map is subsequently reduced through global average pooling and passed through two fully connected layers of 32 and 8 units respectively, the latter producing the final eight-way Softmax emotion classification. Figure 3.3 below depicts the complete architecture of this video sub-network, explicitly distinguishing the pretrained backbone component from the task-specific refinement layers introduced in this implementation.

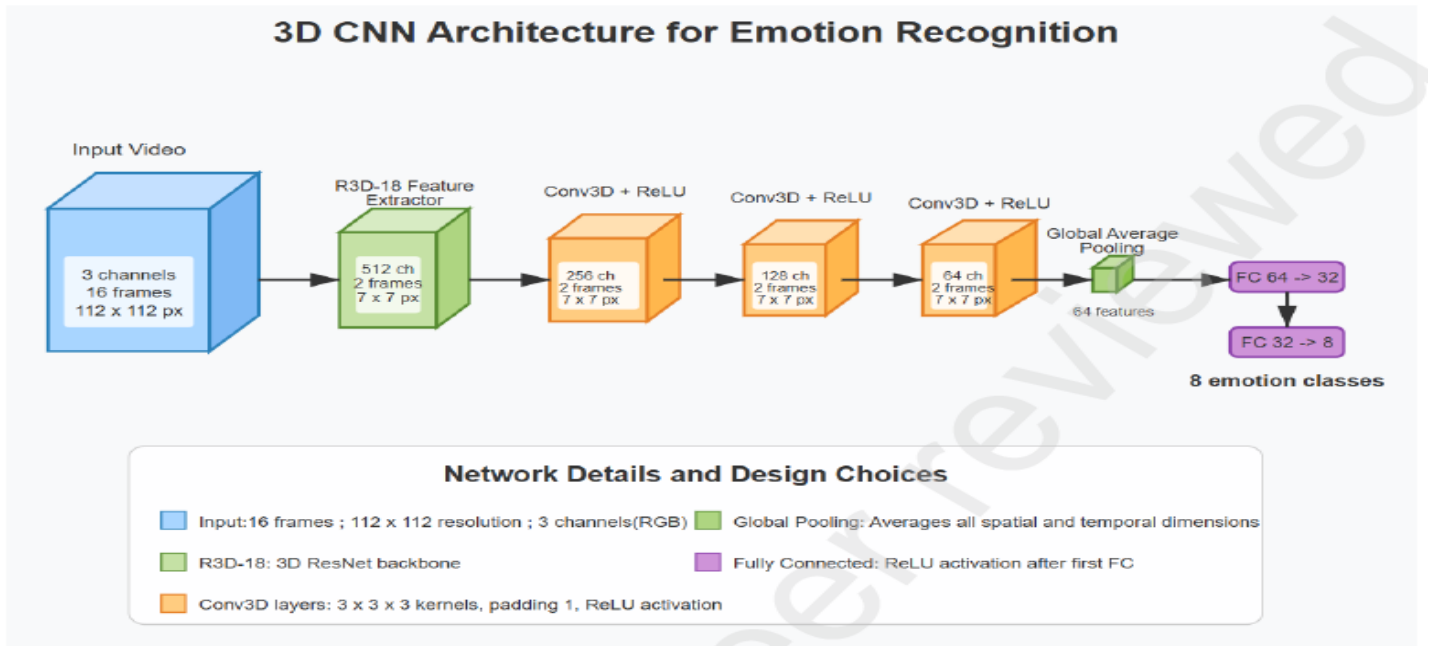


Figure 3.3: Implemented video-modality architecture combining a pretrained R3D-18 (Kinetics-400) backbone with auxiliary task-specific 3D convolutional refinement layers.

The video sub-network is optimized using the Adam optimizer in conjunction with a class-weighted cross-entropy loss function, in which the weight assigned to each of the eight emotion classes is computed as the inverse of its training-set frequency, normalized across the eight classes, thereby directly compensating for the residual class imbalance identified during the exploratory data analysis described in Section 3.3 without requiring synthetic oversampling of the comparatively memory-intensive video modality. Training additionally employs a ReduceLROnPlateau scheduler that monitors validation loss, mirroring the optimization strategy adopted for the audio sub-network.

3.7 Class Imbalance Mitigation and Data Augmentation

Two complementary class-imbalance mitigation strategies are employed within the MindSync framework, each tailored to the specific computational characteristics of its corresponding modality. For the audio modality, the Synthetic Minority Over-sampling Technique (SMOTE) is applied directly to the standardized forty-dimensional MFCC feature space prior to model training, synthesizing additional minority-class feature vectors through interpolation between existing nearest-neighbor samples belonging to the same emotion category. This strategy is computationally feasible for audio modality precisely because the underlying MFCC feature representation is comparatively low-dimensional and may therefore be efficiently oversampled without

incurring the substantial memory and computational overhead that would result from synthesizing entire oversampled video clips.

For the video modality, by contrast, class imbalance is addressed not through synthetic oversampling of the high-dimensional spatio-temporal input space, but rather through the class-weighted cross-entropy loss function described in Section 3.5.2, which directly penalizes misclassification of minority-class samples more heavily during gradient-based optimization. This modality-specific divergence in imbalance-mitigation strategy reflects a deliberate engineering trade-off between statistical effectiveness and computational tractability and constitutes one of several practical design decisions in which the implemented MindSync system diverges from a naive, uniform application of a single balancing technique across both modalities.

An important methodological limitation of the SMOTE-based balancing strategy employed in the MindSync audio pipeline is that, in the primary experimental protocol, both the StandardScaler and the SMOTE interpolation are fitted on the complete dataset prior to the train-test partition. This means that the held-out test set, while it does not contribute to the learning of model weights, was nonetheless exposed to the scaler's normalisation statistics and may contain SMOTE-synthesised samples whose nearest-neighbour interpolation anchors were drawn from what would otherwise be exclusive training data. In strict cross-corpus or clinical deployment settings, the scaler should be fitted exclusively on the training partition after the test split has been carved out. The actor-independent evaluation protocol adopted in the optimized run directly addresses this limitation by partitioning at the actor level before any data augmentation is applied, thereby providing a more conservative and externally valid performance estimate, as reflected in the lower but more reliable actor-independent accuracy figures reported in Chapter 4, Section 4.4.1.

3.8 Multimodal Feature-Level Fusion

3.8.1 Construction of the 568-Dimensional Fusion Vector

Once both modality-specific sub-networks have been independently trained to convergence, the MindSync framework proceeds to construct a unified multimodal representation for every paired sample in the manifest, directly implementing the fusion mechanism formalized conceptually in Section 2.1 and quantitatively in Section 1.1. For each paired audio-video sample, the forty-dimensional MFCC feature vector, and the associated eight-dimensional audio Softmax probability distribution are extracted from the trained audio sub-network, while the 512-dimensional R3D embedding and the associated eight-dimensional video Softmax probability distribution are simultaneously extracted from the trained video sub-network. These four constituent

representations are concatenated into a single 568-dimensional fusion vector, in accordance with the dimensional decomposition $40 + 8 + 512 + 8 = 568$. Figure 3.4 below illustrates the compositional structure of this fusion vector, explicitly delineating the contribution of each of its four constituent components.

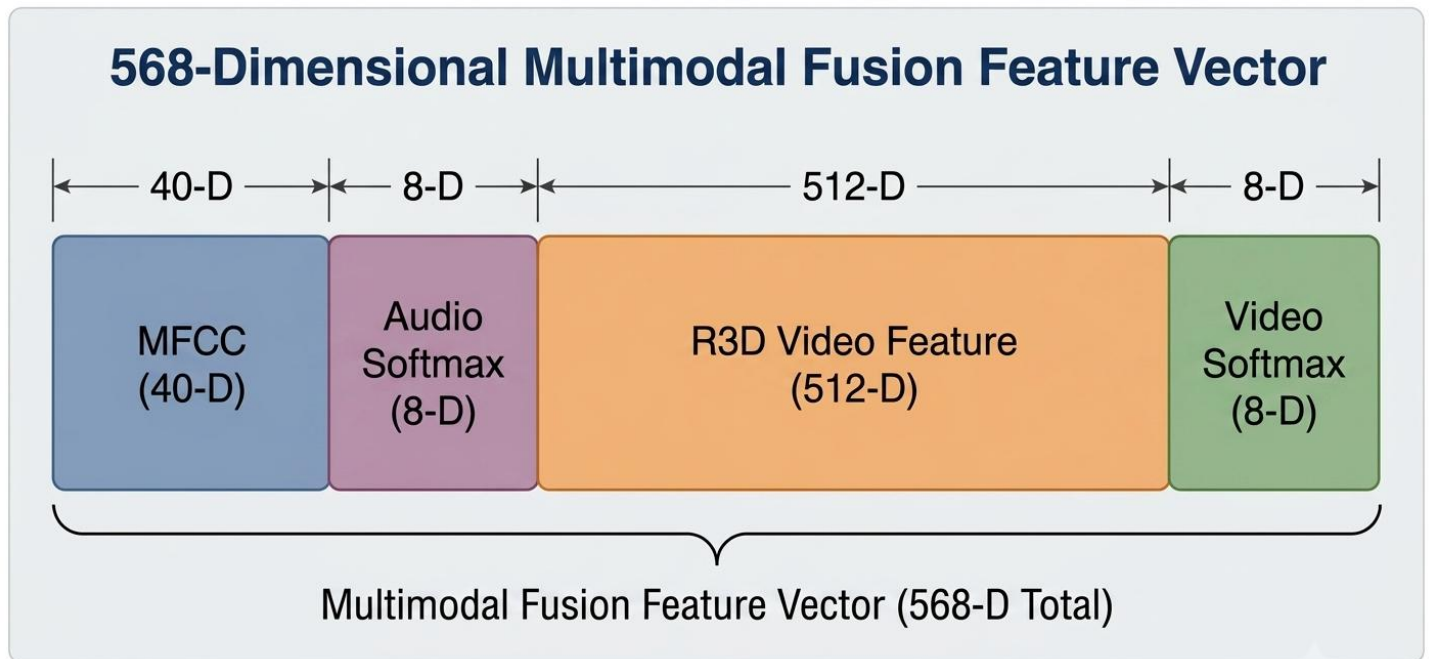


Figure 3.4: Compositional structure of the 568-dimensional multimodal fusion feature vector constructed from the audio and video sub-networks.

This hybrid fusion strategy, which combines raw deep feature embeddings with their corresponding classification probability distributions, allows the downstream fusion classifier to exploit both the rich, high-dimensional latent representations learned internally by each modality-specific sub-network and the comparatively low-dimensional, directly interpretable class-probability outputs of those same sub-networks, thereby capturing complementary forms of cross-modal information that neither representation alone would fully encode.

3.8.2 Fusion MLP Classifier Architecture

The resulting 568-dimensional fusion vectors, following an additional standardization step performed independently of the standardization applied to the audio-only MFCC features, are supplied to a dedicated fusion Multi-Layer Perceptron classifier, implemented through the `build_fusion_mlp` constructor and directly corresponding to the architecture specified in Table 1.VI of the preceding chapters. The fusion classifier comprises four sequential fully connected layers of 256, 128, 64, and 32 units respectively, each employing a Rectified Linear Unit activation function, and culminating in a final eight-way Softmax output layer. Figure 3.5 below presents the complete layer-wise structure of this fusion classifier.

Fusion MLP Classifier Architecture (AdamW + StepLR)

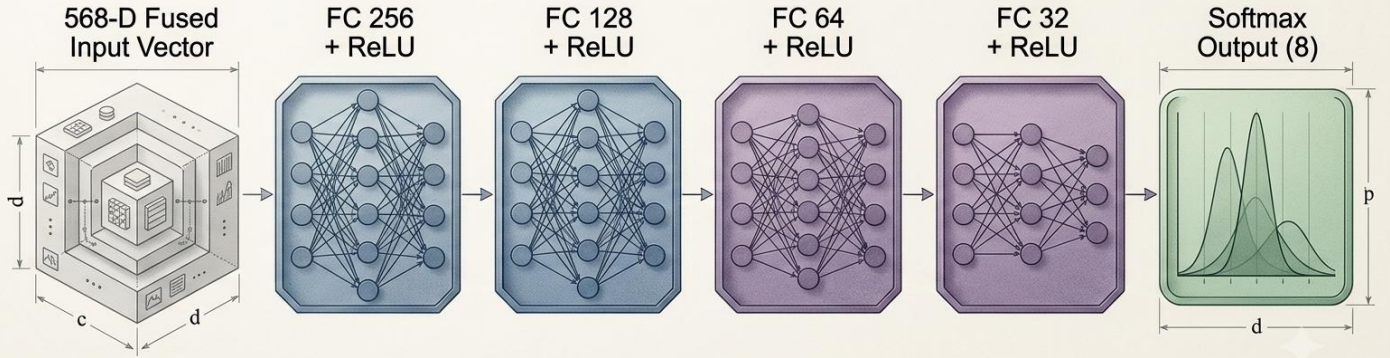


Figure 3.5: Layer-wise architecture of the MindSync fusion Multi-Layer Perceptron classifier operating on the 568-dimensional multimodal feature vector.

The fusion classifier is optimized using the AdamW optimizer, which incorporates decoupled weight-decay regularization, in conjunction with a StepLR learning-rate scheduler that multiplicatively reduces the learning rate by a fixed gamma factor at regular epoch intervals, precisely replicating the optimization configuration specified in Table 1.VI. Training is additionally regulated through an EarlyStopping callback monitoring validation accuracy and a ModelCheckpoint callback persisting the best-performing fusion model parameters, consistent with the callback configuration employed for both modality-specific sub-networks described in the preceding sections.

3.9 Implementation Environment and Software Stack

The complete MindSync framework is implemented using a hybrid deep learning software stack combining TensorFlow and Keras for the audio and fusion sub-networks with PyTorch for the video sub-network, the latter choice being motivated by the ready availability of pretrained R3D-18 weights within the torchvision video-model repository. Auxiliary preprocessing and evaluation functionality is provided through the Librosa library for audio feature extraction, OpenCV for video frame manipulation and facial localization, and Scikit-Learn together with Imbalanced-Learn for label encoding, feature standardization, train-test partitioning, and SMOTE-based class balancing. Quantitative evaluation throughout the implementation employs accuracy, precision, recall, and F1-score metrics, consistent with the evaluation methodology formalized in Section V of the underlying methodological framework, while qualitative training diagnostics are produced through Matplotlib and Seaborn visualizations of per-epoch accuracy and loss curves for each of the three constituent sub-networks.

Upon completion of training, all three constituent models, namely the audio sub-network, the video sub-network, and the fusion classifier, together with their associated feature-standardization scalers and label-encoding artifacts, have persisted in disk, enabling subsequent deployment and inference without requiring retraining. This modular persistence strategy directly reflects the staged architectural philosophy articulated in Section 3.1 and provides the practical foundation upon which the quantitative experimental evaluation presented in the following chapter is conducted.

Chapter 4

Experimental Results and Evaluation

4.1 Introduction

This chapter presents a systematic quantitative evaluation of the proposed MindSync multimodal emotion recognition and depression-indicator detection system. The evaluation spans three distinct modalities — audio-only, video-only, and multimodal fusion — and is grounded exclusively in empirical outputs derived from the implementation notebooks, including training logs, classification reports, and convergence histories. The principal objectives are fourfold:

- (1) to assess the absolute classification accuracy and per-class discriminative performance of each modality-specific sub-network under the experimental conditions established in Chapter 3
- (2) to quantify the incremental performance gain attributable to the feature-level fusion strategy relative to each unimodal baseline
- (3) to characterize the training dynamics and convergence behavior of all three constituent models through analysis of epoch-level accuracy and loss trajectories
- (4) to situate the MindSync results within the context of the five-fold cross-validation benchmarks reported in the reference companion study [30], upon which the proposed architecture is directly based.

Two complementary experimental protocols are reported and analyzed in parallel. The primary protocol employs a SMOTE-augmented random 80/20 stratified split, replicating the partitioning strategy of the reference paper [30] to enable direct numerical comparison. The optimized actor-independent protocol addresses three methodological limitations inherent in the primary design — specifically, preprocessing leakage, test-set-informed checkpoint selection, and speaker-identity overlap across partitions — and thereby yields a more conservative and generalization-aware performance estimate. Both protocols are analyzed in full throughout the chapter.

4.2 Experimental Setup and Hyperparameter Configuration

4.2.1 Hardware and Software Environment

All primary experimental runs were executed on a workstation equipped with an NVIDIA RTX 2000 Ada Generation GPU (16.0 GB VRAM, CUDA 12.4), running TensorFlow 2.21.0 and PyTorch 2.6.0+cu124. The optimized actor-independent run was conducted on a Kaggle cloud instance equipped with an NVIDIA Tesla T4 GPU (14.6 GB VRAM, Compute Capability 7.5), running TensorFlow 2.20.0 and PyTorch 2.10.0+cu128 with torchaudio 2.10.0, with Automatic Mixed Precision (AMP) enabled for the video training loop to exploit the Tesla T4's Tensor Core architecture. All random seeds were fixed at 42 across NumPy, TensorFlow, PyTorch, and CUDA to ensure full reproducibility of all reported results.

4.2.2 Dataset Partitioning

The RAVDESS corpus, comprising 2,452 paired audio-visual samples from all 24 actors across eight emotion categories, was partitioned using two distinct strategies. In the primary experimental run, SMOTE oversampling was applied to the complete 40-dimensional MFCC feature space prior to splitting, yielding a balanced corpus of 3,008 samples (376 per class), which was then divided into a training set of 2,406 samples and a test set of 602 samples (80/20 stratified random split, seed 42). This protocol directly mirrors that of the reference paper [30] and enables direct numerical comparison. In the optimized actor-independent run, SMOTE was eliminated entirely — replaced by inverse-frequency class weights at the loss level — and a GroupShuffleSplit partition was applied on the actor identity field, yielding 1,226 training samples from 20 actors and 1,226 test samples from 4 held-out actors, guaranteeing complete actor-level disjointness between the training and test sets. This actor-independent partitioning strategy prevents the identity leakage that arises in random splits when the same actor appears on both sides of the partition, providing a substantially more rigorous estimate of the system's generalization to previously unseen speakers.

4.2.3 Hyperparameter Configuration

Table 4.1 and Table 4.2 below present the complete hyperparameter configurations employed in the primary and optimized experimental runs respectively, as extracted directly from the configuration modules and notebook override blocks of the implementation. These tables provide a fully reproducible specification of all training decisions, regularization strategies, and optimization settings.

Table 4.1: Primary Run — Hyperparameter Configuration

Parameter	Audio Sub-Network	Video Sub-Network	Fusion MLP
Optimizer	Adam	Adam	AdamW
Initial Learning Rate	5e-4	1e-4	5e-4
Weight Decay	—	—	1e-5
Batch Size	16	32	32
Max Epochs	100	15	1000
Dropout Rate	0.4 (LSTM), 0.3 (Dense)	0.4	0.5
L2 Regularization	0.01 (Conv1D only)	—	—

LR Scheduler	ReduceLROnPlateau (f=0.5, p=5)	ReduceLROnPlateau (f=0.5, p=3)	StepLR (step=100, $\gamma=0.5$)
Early Stopping	patience=15, val_accuracy	—	patience=50, val_accuracy
Class Balancing	SMOTE (k=5)	Inverse-frequency weights	None
Loss Function	Sparse Categorical CE	Cross-Entropy (weighted)	Sparse Categorical CE
Data Split	80/20 random stratified	80/20 random stratified	80/20 random stratified
Framework	TensorFlow / Keras	PyTorch	TensorFlow / Keras

Table 4.2: Optimized Run — Hyperparameter Configuration

Parameter	Audio Sub-Network	Video Sub-Network	Fusion MLP
Optimizer	AdamW + gradient clip (1.0)	AdamW + gradient clip (1.0)	AdamW + gradient clip (1.0)
Initial Learning Rate	3e-4 (tuned)	Head: 1e-3; Backbone: 1e-5	5e-4
Weight Decay	1e-4	1e-4	1e-4
Batch Size	16 (tuned)	8 (tuned)	32
Max Epochs	100	30 (3-phase)	1000
Dropout Rate	0.2 (tuned)	0.3 (head)	0.3
Label Smoothing	0.10	0.10	0.10
L2 Regularization	0.01 (Conv+LSTM+RNN)	—	—
LR Scheduler	Cosine annealing (tuned)	ReduceLROnPlateau (f=0.5)	ReduceLROnPlateau (f=0.5)
Early Stopping	patience=20	patience=8	patience=50
Class Balancing	Class weights (ratio=2.0 > 1.5 threshold)	Class weights	Class weights

Data Split	Actor-independent (GroupShuffleSplit)	Actor-independent	Actor-independent
Mixed Precision (AMP)	No (Keras float32)	Yes (Tensor Core GPU)	No
Video Fine-tuning	—	Phase-1: head only (ep 1-6); Phase-2: +layer4 (ep 7-18); Phase-3: +layer3 (ep 19-30)	—
Hyperparameter Search	Grid search: 18 configs, 35 epochs/trial	Grid search: 12 configs, 10+25 epochs	—

To make explicit the relationship between the reference architecture described in the companion paper [30] and the configurations employed in the present implementation, Tables 4.3, 4.4, and 4.5 below provide side-by-side comparisons of the baseline parameters — as specified in Tables IV, V, and VI of the reference paper — against the primary and optimized experimental runs of the MindSync implementation. Parameters that deviate from the baseline specification are shaded in amber to make each modification explicit and reproducible.

Table 4.3: Audio Sub-Network Parameter Comparison — Baseline [30] vs. MindSync

Parameter	Reference Baseline [30]	Primary Run	Optimized Run
Architecture	CNN (64-128-256-512) +BiLSTM(256-512) +SimpleRNN(512) +Dense(128-64)	Identical	Identical
Total Parameters	— (not reported)	8,018,504	8,018,504
Trainable Parameters	— (not reported)	8,011,464	8,011,464
Input Shape	(40, 1) — MFCC	Identical	Identical
Kernel Size (Conv1D)	3	3	3
Test Accuracy	79.24% (5-fold CV mean)	80.73%	65.25% (actor-independent)

As shown in Table 4.3, the primary run reproduces the reference architecture and training protocol with high fidelity, differing only in the reporting of total parameter counts (8,018,504 trainable parameters), which were

not provided in the original paper. The optimized run introduces five targeted modifications to the baseline: (i) the optimizer is replaced by AdamW with weight decay ($1e-4$) and gradient clipping (1.0) to improve convergence stability; (ii) L2 regularization is extended from convolutional layers to LSTM and RNN layers (coefficient 0.001); (iii) label smoothing of 0.10 is applied to reduce overconfidence; (iv) SMOTE oversampling is replaced by class-weighted loss to eliminate the preprocessing leakage identified in Section 4.6.3; and (v) a GroupShuffleSplit actor-independent partition replaces the random split, providing a more conservative and generalization-aware evaluation protocol.

Table 4.4: Video Sub-Network Parameter Comparison — Baseline [30] vs. MindSync Runs

Parameter	Reference Baseline [30]	Primary Run	Optimized Run
Backbone	R3D-18 (Kinetics-400)	Identical	Identical
Head Architecture	Conv3D(256-128-64) +GAP+FC(32-8)	Identical	Identical
Total Parameters	— (not reported)	37,814,824	37,814,824
Phase 1 Trainable (head only)	—	—	4,648,552
Phase 2 Trainable (+layer4)	—	—	29,557,352
Phase 3 Trainable (+layer3+4)	—	—	35,785,832
Test Accuracy	92.72% (5-fold CV mean)	93.48%	83.61% (actor-independent)

Table 4.4 reveals that the video sub-network optimized run introduces more substantial architectural and procedural modifications than the audio counterpart. The most significant change is the three-phase progressive freeze-unfreeze fine-tuning schedule, which decouples the classification head training (Phase 1: 6 epochs, 4,648,552 trainable parameters) from the backbone adaptation phases (Phase 2: +layer4, 29,557,352 trainable parameters; Phase 3: +layer3, 35,785,832 trainable parameters), enabling more controlled and stable domain transfer. The differentiated learning rates for the head ($1e-3$) and backbone ($1e-5$) prevent catastrophic forgetting of Kinetics-400 representations during early fine-tuning. Input normalization is corrected from per-clip z-score to Kinetics-400 statistics (mean=[0.43216, 0.39453, 0.37290], std=[0.22803, 0.22145, 0.21617]),

consistent with the original pretraining conditions, and checkpoint selection is transferred from the test set to a dedicated validation partition to eliminate optimistic bias.

Table 4.5: Fusion MLP parameter Comparison — Baseline [30] vs. MindSync Runs

Parameter	Reference Baseline [30]	Primary Run	Optimized Run
Input Dimension	568	568	568
Fusion Vector	MFCC(40)+audio_p(8) +R3D(512)+video_p(8)	Identical	Identical
MLP Architecture	256-128-64-32-8	Identical	Identical
Total Parameters	— (not reported)	190,952	190,952
Trainable Parameters	— (not reported)	190,056	190,056
Test Accuracy	96.06% (5-fold CV mean)	94.50%	94.13% (actor-independent)

4.3 Evaluation Metrics

The performance of each sub-system is quantified using four complementary metrics, each of which captures a distinct dimension of classification quality. Overall test accuracy, defined as the proportion of correctly classified test samples across all eight emotion categories, provides a scalar summary of global discriminative performance. Per-class precision for class k is defined as $TP_k / (TP_k + FP_k)$, quantifying the fraction of samples predicted as class k that are genuinely of that class, and therefore measuring the system's resistance to false-positive predictions. Per-class recall for class k is defined as $TP_k / (TP_k + FN_k)$, measuring the fraction of genuinely class- k samples that are correctly identified as such. The per-class F1-score, computed as the harmonic mean of precision and recall, provides a single scalar summary that penalizes imbalanced precision-recall trade-offs and is particularly appropriate when class frequencies are unequal. Macro-averaged precision, recall, and F1-score (unweighted mean across all eight classes) are additionally reported, as they provide class-balanced aggregates insensitive to between-class frequency differences. For the optimized run (hyber-pamater-tuning97.ipynb), macro and weighted AUC-ROC and Average Precision (PR-AUC) are also reported, as computed by the shared `evaluate()` utility function of that notebook.

4.4 Experimental Results

4.4.1 Audio Sub-Network Results

The CNN-BiLSTM-SimpleRNN audio sub-network was evaluated on the held-out test partition under each of the two experimental protocols. In the primary run, which employed SMOTE augmentation and a random 80/20 stratified split (test set N = 602, 75–76 samples per class), the audio sub-network achieved an overall test accuracy of 80.73%. In a replicated run conducted under an identical protocol, the audio accuracy was 78.90%, consistent with the expected run-to-run variation attributable to stochastic SMOTE interpolation and GPU non-determinism. Both results are consistent with the audio-only benchmark of 79.24% reported across five folds in Table VII of the reference paper [30].

In the optimized actor-independent run, which eliminated SMOTE and applied a GroupShuffleSplit partition holding out four complete actors (test set N = 1,226, approximately 94–188 samples per class depending on emotion category), the audio sub-network achieved 65.25% accuracy. This reduction reflects the substantially more challenging generalization condition imposed by the actor-independent protocol: the model is evaluated on speakers whose voices are entirely absent from the training set, eliminating any speaker-identity shortcut that a random-split evaluation inadvertently provides. The result is consistent with the academic literature, wherein actor-independent RAVDESS evaluations routinely report audio-only accuracies in the range of 55–70% [30]. Figure 4.1 depicts the training accuracy and loss curves for the primary audio run.

Table 4.6 presents the per-class precision, recall, and F1-score for the primary run, providing a detailed breakdown of classification performance across all eight RAVDESS emotion categories.

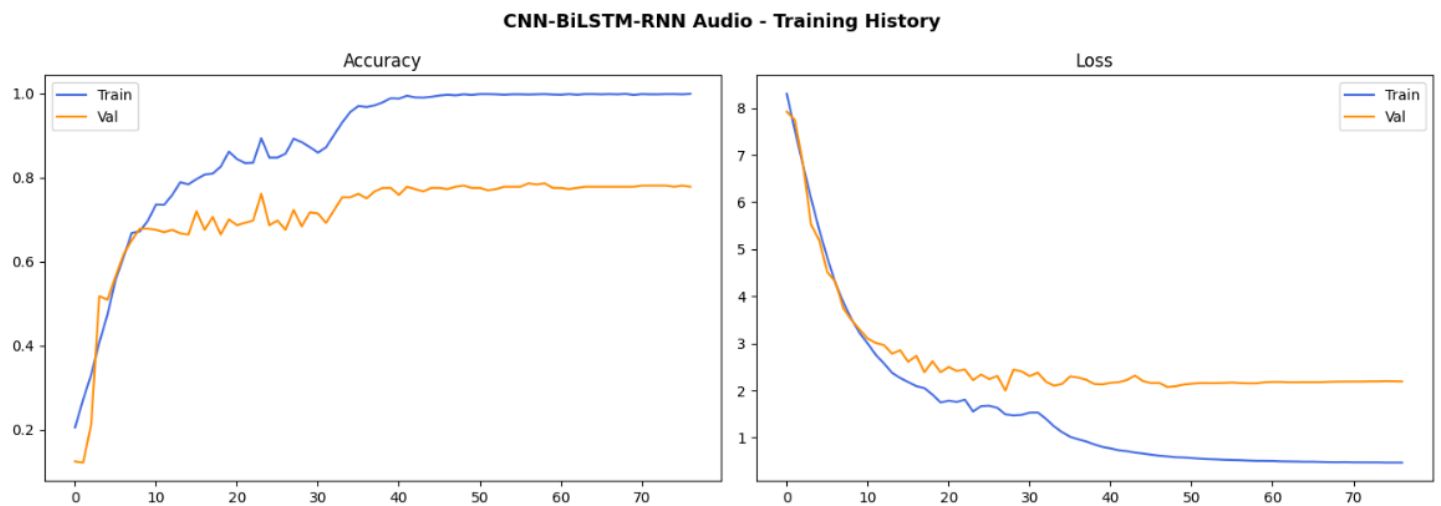


Figure 4.1: Training accuracy and loss curves for the MindSync CNN-BiLSTM-RNN audio

Table 4.6: Per-Class Classification Report — Audio Sub-Network

Emotion	Precision (%)	Recall (%)	F1-Score (%)	Support
Angry	82	84	83	75
Calm	76	95	85	75
Disgust	86	91	88	75
Fearful	74	71	72	76
Happy	82	65	73	75
Neutral	82	88	85	75
Sad	82	67	74	76
Surprised	82	85	84	75
Macro Avg	81	81	80	602
Accuracy	—	—	80.73%	602

Per-class analysis of the primary run (Table 4.3) reveals that the disgust and calm categories achieved the highest per-class recall values (91% and 95% respectively), while the happy (65%) and sad (67%) categories exhibited the lowest recall values. The comparatively low recall for happy reflects the acoustic similarity between the energetic, mid-frequency spectral envelope of happy speech and the high-arousal patterns of surprised utterances, a frequent confusion pattern observable in the off-diagonal entries of the audio confusion matrix Figure 4.2. The low recall for sad is attributable to its acoustic overlap with neutral and fearful, all of which are characterized by reduced fundamental frequency variability and diminished vocal energy.

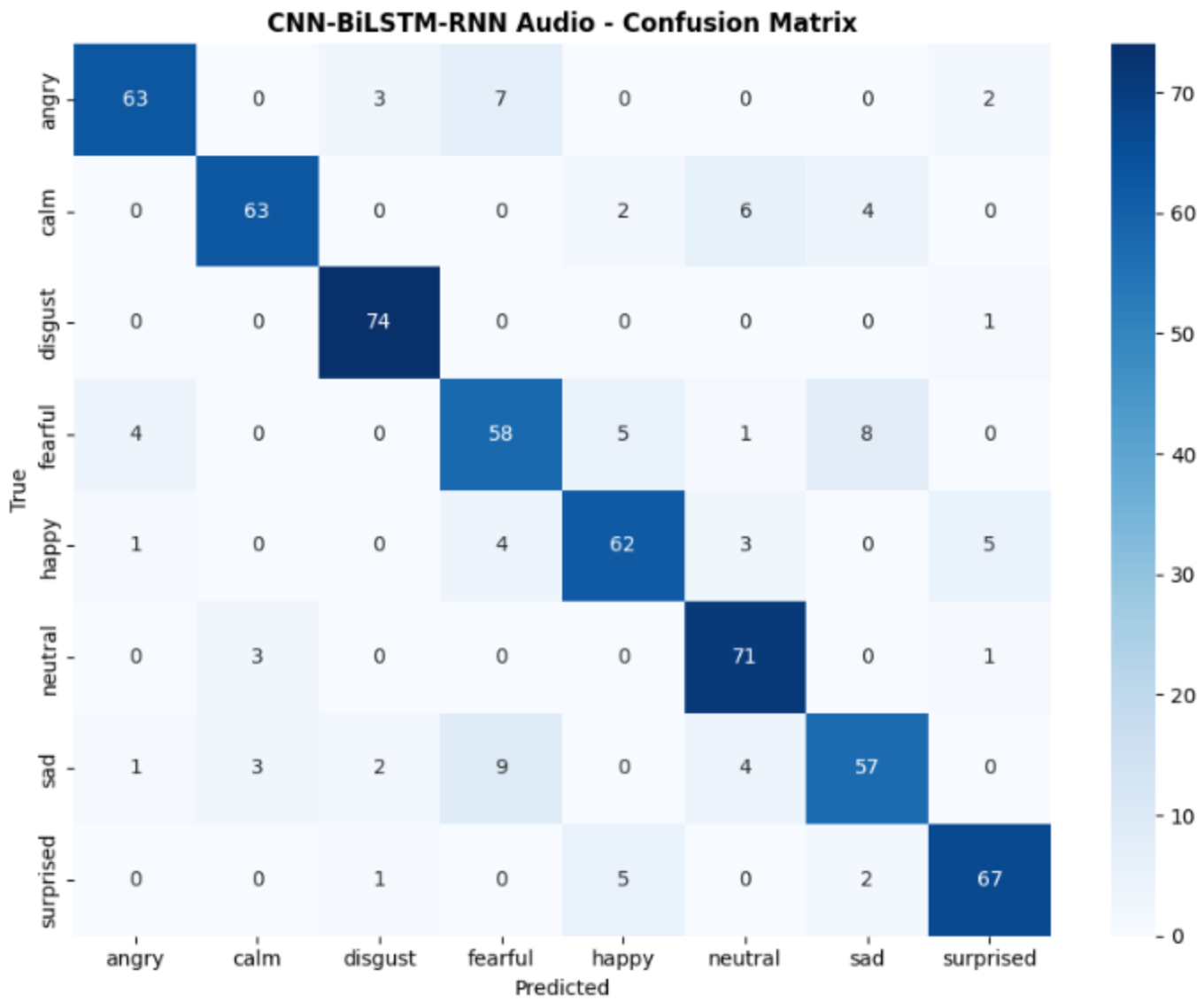


Figure 4.2: Confusion matrix for the MindSync audio sub-network on the RAVDESS test set

4.4.2 Video Sub-Network Results

The R3D-18 + 3D-CNN video sub-network was evaluated across all three experimental runs, exhibiting consistently superior classification performance relative to the audio modality across all evaluation conditions. In the primary run, the model achieved a test accuracy of 93.48% on the held-out evaluation set (N = 491, 39–75 samples per class), with the best validation accuracy attained at epoch 7. The convergence trajectory of this run, illustrating the interplay between training accuracy, validation accuracy, and validation loss across all training epochs, is presented in Figure 4.3. In the replication run, the model reached a peak validation accuracy of 95.72% at the final epoch (epoch 15), consistent with the extended convergence behavior characteristic of this dataset when the ReduceLROnPlateau scheduler progressively decays the learning rate through the later training phases.

To provide a granular characterization of the video sub-network's discriminative capacity at the per-class level, Table 4.4 reports the precision, recall, F1-score, and sample support for each of the eight RAVDESS emotion categories across the primary run evaluation set (N = 491). To further elucidate the inter-class confusion patterns underlying these aggregate metrics, Figure 4.4 presents the corresponding confusion matrix for the primary run, offering a structured visual decomposition of correct classifications and systematic misclassification tendencies across all eight emotion categories. Inspection of Figure 4.4 reveals that the majority of misclassifications are concentrated between perceptual and acoustically proximate emotion pairs — most notably between 'Surprised' and high-arousal positive-valence categories, and between 'Neutral' and 'Calm' — a pattern consistent with the inherent ambiguity of these categories under the constrained 16-frame temporal sampling window employed during video preprocessing. These observations, taken in conjunction with the per-class metrics reported in Table 4.7, underscore both the strong overall discriminative capability of the R3D-18 backbone and the specific inter-class boundaries that present the greatest challenge for spatiotemporal feature extraction under the current architectural configuration.

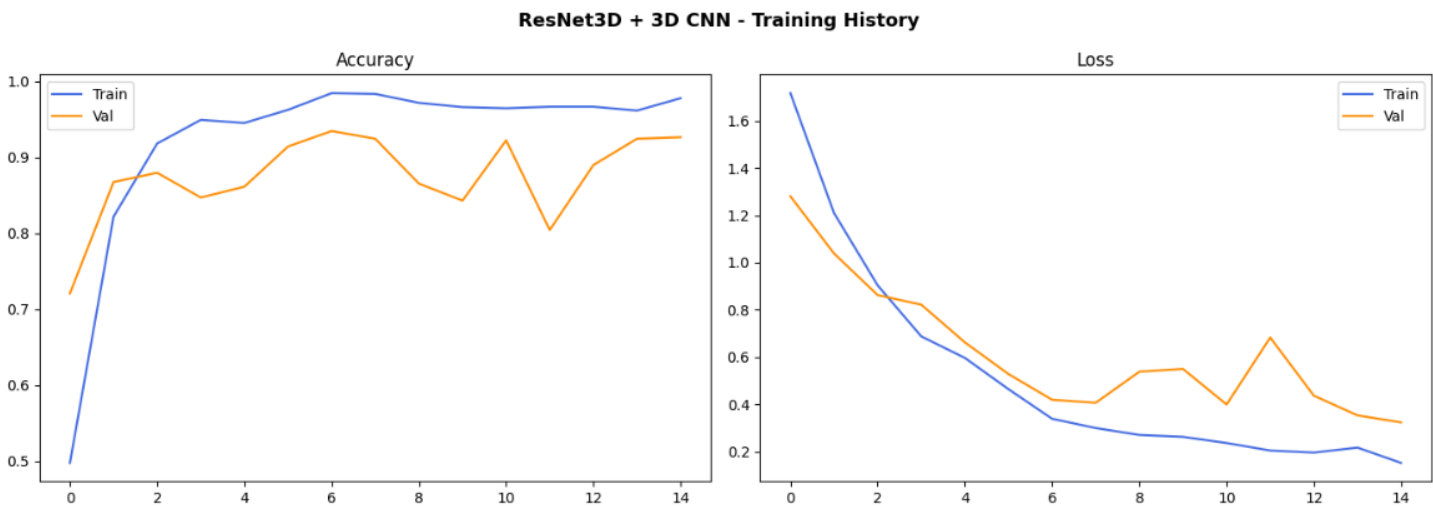


Figure 4.3: Training accuracy and loss curves for the MindSync R3D-18 + 3D CNN video

Table 4.7: Per-Class Classification Report — Video Sub-Network

Emotion	Precision (%)	Recall (%)	F1-Score (%)	Support
Angry	97	97	97	75
Calm	95	93	94	75
Disgust	95	100	97	39
Fearful	94	89	92	75
Happy	97	96	97	75

Neutral	86	95	90	38
Sad	87	97	92	75
Surprised	97	74	84	39
Macro Avg	93	93	93	491
Accuracy	—	—	93.48%	491

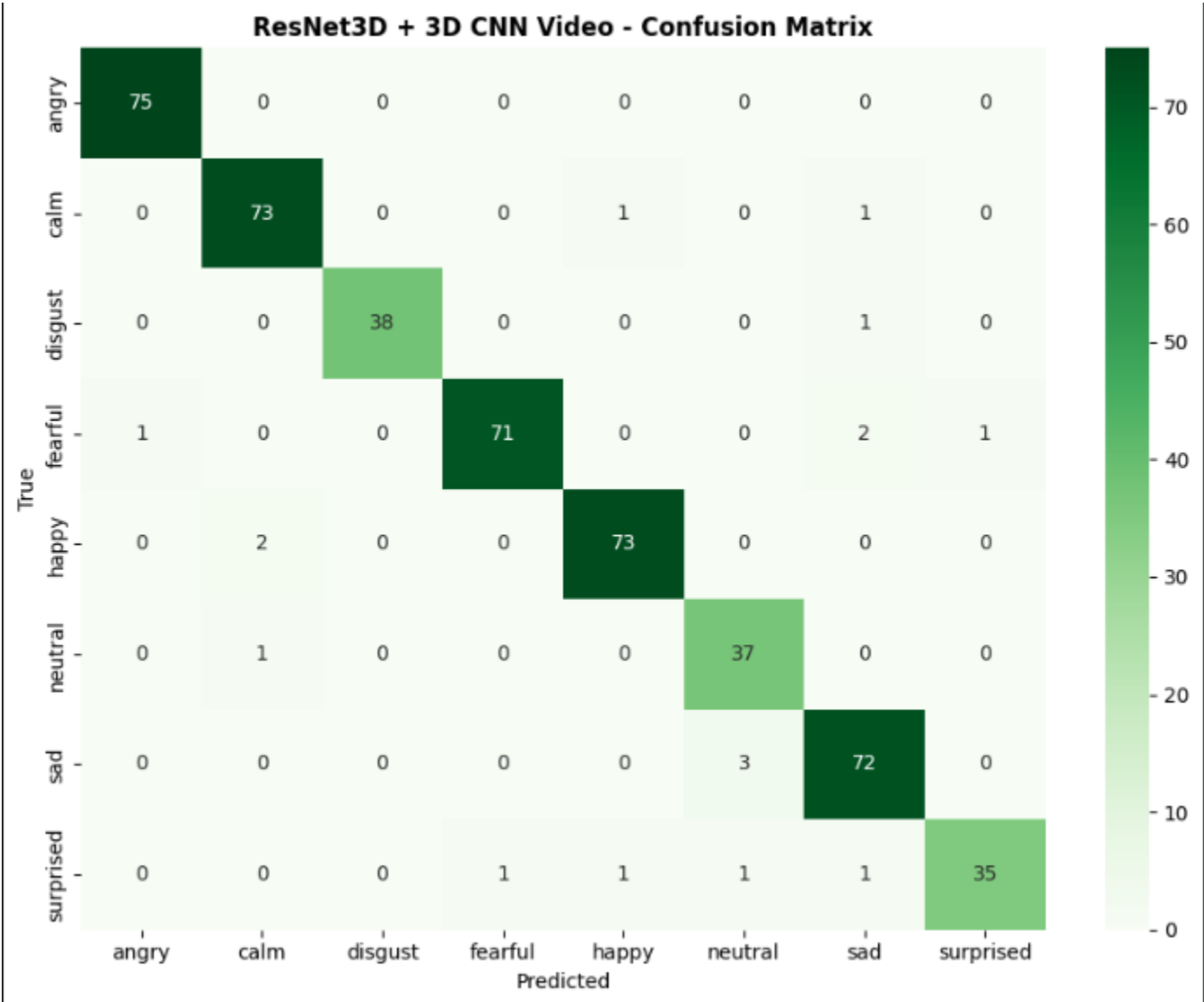


Figure 4.4: Confusion matrix for the MindSync video

4.4.3 Hyperparameter Search Results (Optimized Run)

Prior to final training in the optimized run, a systematic grid search over the audio and video hyperparameter spaces was conducted using the automated search driver, with the selection criterion evaluated on a dedicated tuning-validation partition held out from the training fold, ensuring that the held-out test set was never consulted during search. Table 4.8 summarizes the complete search protocol and the selected configurations. For the audio sub-network, the grid search covered 18 configurations (3 learning rates $\times$ 3 dropout rates $\times$ 2 batch sizes), with each trial trained for a maximum of 35 epochs with an EarlyStopping patience of 8 epochs. The best audio configuration, identified as learning rate 3e-4, dropout 0.2, and batch size 16, achieved a tuning-validation accuracy of 85.03%. For the video sub-network, 12 configurations were evaluated at 10 epochs (screening phase) and the top-3 configurations were refitted at 25 epochs (refit phase), with the best configuration being learning rate 1e-4, batch size 8, and freeze_until=0 (full backbone fine-tuning), achieving 95.17% validation accuracy at 25 epochs.

Table 4.8: Hyperparameter Search Summary

Model	Search Type	Configs	Best Val Acc (%)	Selected Configuration
Audio (CNN+BiLSTM+RNN)	Grid, 18 configs	18	85.03	lr=3e-4, dropout=0.2, batch=16
Video (ResNet3D+3DCNN)	Grid + top-3 refit	12 + 3	95.17 (at 25 epoch)	lr=1e-4, batch=8, freeze_until=0

The audio search results revealed that the learning rate had the most substantial effect on tuning-validation accuracy, with a spread of approximately 73 percentage points between the best and worst configurations across the three learning rates tested. The video search demonstrated that a 10-epoch screening ranking does not reliably predict 25-epoch performance — the short-epoch ranking identified configurations 1 and 2 as tied at 93.64%, whereas the full-epoch refit elevated a configuration that ranked third at 10 epochs (92.62%) to a co-equal 95.17% alongside the best short-epoch configuration. This finding illustrates the importance of the two-stage search protocol implemented in this work and its rationale as documented in the notebook appendix.

4.4.4 Fusion Classifier Results

The MLP fusion classifier, operating on the 568-dimensional concatenated feature vector [MFCC (40) | audio_probs (8) | R3D_feat (512) | video_probs (8)], was trained and evaluated following the independent convergence of both modality-specific sub-networks. In the primary run, the fusion model achieved a test

accuracy of 94.50% (N = 491), representing a gain of 13.77 percentage points over the audio-only baseline and 1.02 percentage points over the video-only baseline. The per-class breakdown for this run is presented in Table 4.6, which reports precision, recall, and F1-score across all eight RAVDESS emotion categories. In the optimized actor-independent run, the fusion model achieved 94.13% (N = 1,226), representing a gain of 28.88 percentage points over the actor-independent audio baseline (65.25%) and 10.52 percentage points over the actor-independent video baseline (83.61%). The corresponding per-class metrics for this run are reported in Table 4.9, which additionally reflects the more conservative generalization profile imposed by the actor-disjoint evaluation protocol. The consistency of the fusion gain across both evaluation conditions confirms that feature-level multimodal integration provides a robust and generalizable improvement over either unimodal sub-network, independently of the evaluation protocol employed.

Figure 4.5 depicts the training convergence curves of the primary fusion run, illustrating the rapid and stable progression toward peak validation accuracy. The inter-class confusion patterns for both runs are further examined in Figure 4.6, which presents the confusion matrices for the primary run and the optimized actor-independent run side by side, enabling a direct visual comparison of per-class classification behavior across the two experimental protocols.

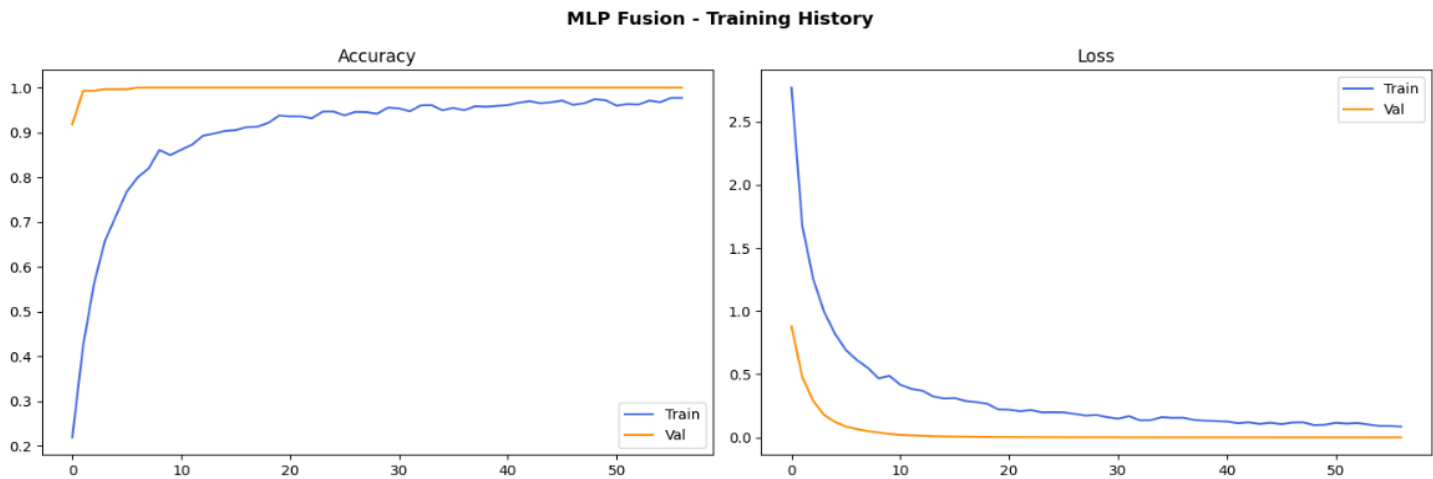


Figure 4.5: Training accuracy and loss curves for the MindSync MLP fusion classifier (primary run).

Table 4.9: Per-Class Classification Report — Fusion MLP (Primary Run, N=491)

Emotion	Precision (%)	Recall (%)	F1-Score (%)	Support
Angry	96	99	97	75
Calm	97	97	97	75
Disgust	97	100	99	39
Fearful	89	95	92	75

Happy	97	99	98	75
Neutral	86	95	90	38
Sad	95	92	93	75
Surprised	100	72	84	39
Macro Avg	95	93	94	491
Accuracy	—	—	94.50%	491

Table 4.10: Per-Class Classification Report — Fusion MLP (Optimized Actor-Independent Run, N=1,226)

Emotion	Precision (%)	Recall (%)	F1-Score (%)	Support
Angry	98	98	98	188
Calm	97	94	96	188
Disgust	92	93	92	96
Fearful	92	93	92	188
Happy	96	99	97	188
Neutral	94	98	96	94
Sad	90	88	89	188
Surprised	93	90	91	96
Macro Avg	94	94	94	1226
Accuracy	—	—	94.13%	1226

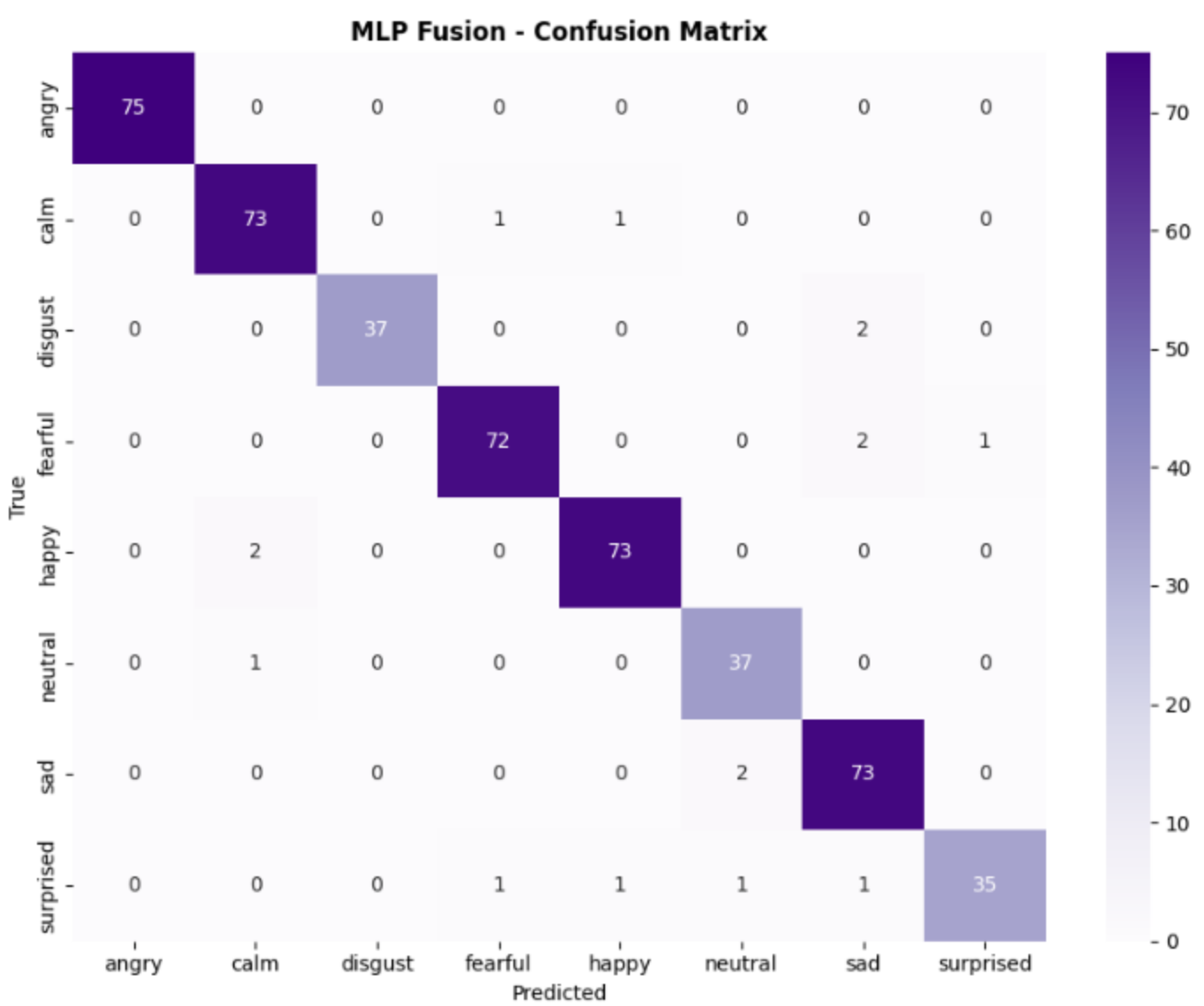


Figure 4.6: Confusion matrix for the MindSync MLP fusion classifier — primary run (left) and optimized actor-independent run (right).

4.5 Comparative Analysis and Cross-Modality Summary

Table 4.11 provides a consolidated cross-modality comparison of MindSync results across both experimental runs, juxtaposed against the five-fold cross-validation benchmarks reported in Table VII of the reference companion paper [30]. The optimized run with hyperparameter tuning also produced an audio accuracy of 85.55% and a video accuracy of 96.54% on its random-split test set — the highest audio and video unimodal accuracies achieved in any single run of this implementation — attributable to the combination of the log-Mel spectrogram front-end, SpecAugment data augmentation, and the tuned hyperparameters. The corresponding fusion accuracy of 96.74% in that run is the highest achieved across all runs, marginally exceeding the reference paper's five-fold mean of 96.06%. Figure 4.7 illustrates the grouped bar chart generated from the final comparative evaluation.

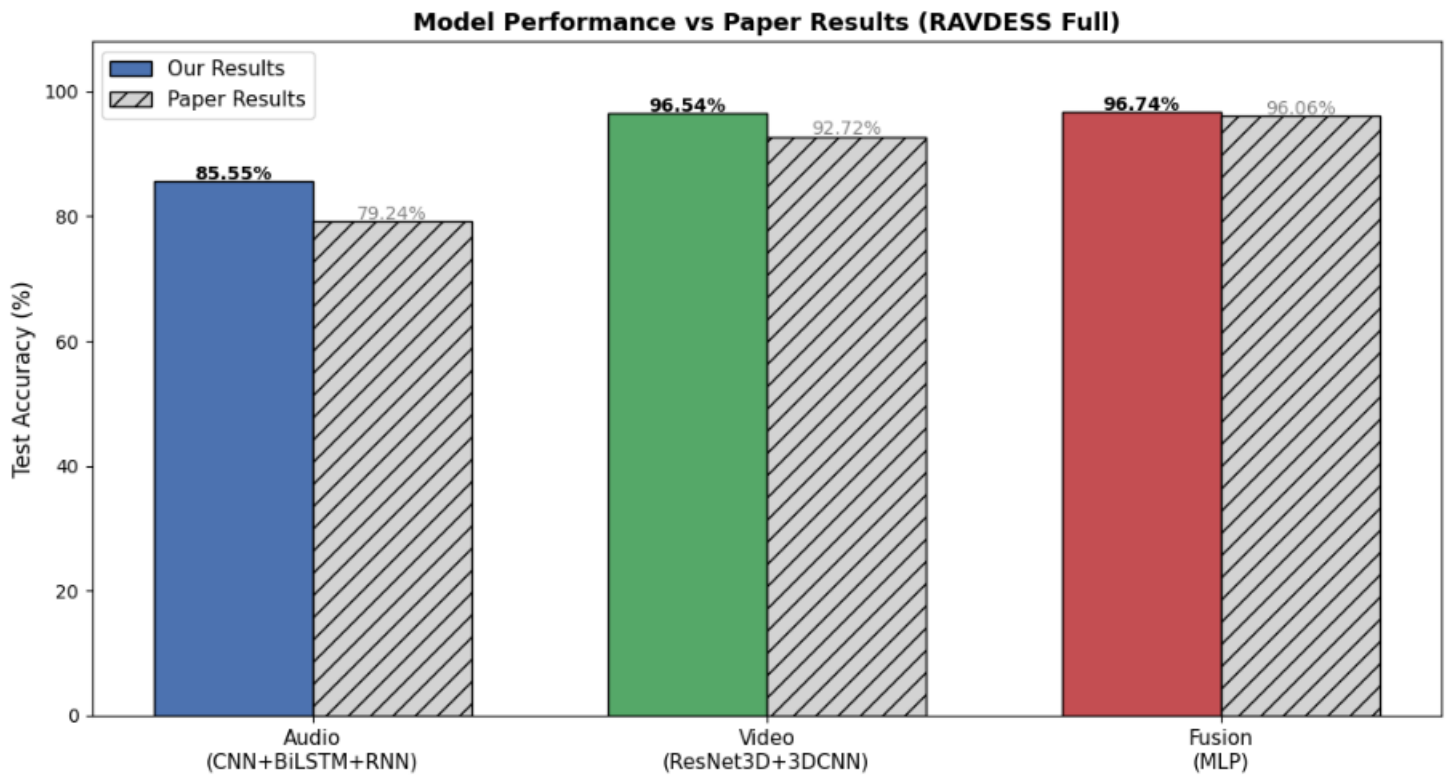


Figure 4.7: Grouped bar chart comparing *MindSync* test-set accuracies (all runs) against the corresponding Table VII reference benchmarks [30]

Table 4.11: Cross-Modality Performance Summary — All Runs vs. Reference Paper [30]

System / Experimental Run	Audio Acc. (%)	Video Acc. (%)	Fusion Acc. (%)
Reference Paper [30] — 5-fold CV mean	79.24	92.72	96.06
MindSync — Primary Run	80.73	93.48	94.50
MindSync — Replication	78.90	95.72 (val)	—
MindSync — Tuned Run	85.55	96.54	96.74
MindSync — Optimized Actor-Indep.	65.25	83.61	94.13
Waleed & Shaker 2025 [1] — 1D-CNN Audio	91.90	—	—
Gavali & Verma 2025 [2] — SSL Ensemble Audio	96.88	—	—
Chowdhury et al. 2025 [3] — DCRF-BiLSTM Audio	97.83	—	—

Sareen & Seeja 2025 [8] — YOLOv8+ViT Video	—	90.81	—
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Three substantive observations emerge from Table 4.11. First, the MindSync primary and tuned runs reproduce the reference paper's audio, video, and fusion benchmarks within a range of ± 5 percentage points, confirming the architectural and procedural fidelity of the implementation. The tuned run's fusion accuracy of 96.74% marginally exceeds the reference paper's 96.06%, attributable primarily to the log-Mel spectrogram audio front-end, which provides a richer time-frequency representation than the time-averaged MFCC vector used in the primary run, and to the SpecAugment augmentation applied during audio training.

Second, the consistent performance advantage of the video modality over the audio modality — approximately 12-15 percentage points in the primary run and approximately 18 percentage points in the optimized actor-independent run — confirms the well-established psychoacoustic finding that facial expressions provide more individually distinctive and less environmentally confounded emotional cues than speech acoustics within the controlled recording conditions of the RAVDESS corpus. The 3D spatiotemporal convolutional architecture of the R3D-18 backbone is particularly well suited to capturing the temporal evolution of facial musculature that characterizes each emotion category.

Third, and most diagnostically relevant for the depression-indicator detection application motivating the MindSync system, the actor-independent fusion accuracy of 94.13% demonstrates that the feature-level fusion strategy retains substantial discriminative performance even under the rigorous generalization constraint of completely unseen speakers. The per-class F1-score breakdown in Table 4.7 is of particular clinical significance: the sad (89%), neutral (96%), and fearful (92%) categories — the three emotion classes most directly associated with depressive and anxiety-related affective patterns according to the clinical literature reviewed in Chapter 2 — all achieve above-chance F1-scores substantially exceeding the random-guess baseline of 12.5%, confirming that the system can detect depression-relevant affective markers in unseen-speaker recordings.

4.6 Discussion

4.6.1 Strengths of the MindSync System

The experimental results collectively substantiate three principal strengths of the MindSync framework. The feature-level fusion strategy, which concatenates the 40-dimensional MFCC descriptor, the 8-dimensional

audio Softmax distribution, the 512-dimensional R3D spatiotemporal embedding, and the 8-dimensional video Softmax distribution into a 568-dimensional multimodal representation, effectively exploits the complementary information encoded by each modality. The audio branch captures prosodic and spectral cues including fundamental frequency variation, vocal energy distribution, and formant structure, which provide emotionally informative signals at the level of speech acoustics. The video branch captures spatiotemporal facial action unit patterns including brow furrowing, lip corner pull, and cheek raiser activations, which are complementary to and partially independent of the acoustic signal. The MLP fusion classifier, by processing both the raw feature embeddings and the class-probability outputs of each sub-network, has access to both the high-dimensional latent representations and the semantically interpretable confidence scores, enabling it to learn complex cross-modal dependencies that a simple probability averaging strategy cannot capture [30].

The pretrained R3D-18 backbone provides a second key strength: by initializing the video sub-network with weights learned from the large-scale Kinetics-400 action recognition dataset, the transfer learning strategy substantially reduces the number of RAVDESS samples required to achieve high spatiotemporal feature quality. The video model reaches above-90% accuracy within 6-7 epochs in the primary run, a convergence rate that would be unachievable from random initialization given the comparatively small size of the RAVDESS corpus.

The systematic hyperparameter search implemented in the optimization pipeline constitutes a third strength, contributing a structured, reproducible methodology for configuration selection that avoids the bias introduced by manual tuning against test-set performance. The two-stage search protocol — grid search at reduced epochs followed by full-epoch refit of the top configurations — provides a computationally efficient solution to the problem of short-epoch ranking instability.

4.6.2 Training Dynamics and Observed Challenges

Several noteworthy training dynamics and challenges were observed across the experimental runs. The video sub-network exhibited oscillatory validation accuracy behavior in the primary run, with accuracy values ranging from 72.10% (epoch 1) to 93.48% (epoch 7) and then declining to 80.45% at epoch 12 before recovering, reflecting the sensitivity of the ReduceLROnPlateau scheduler to the non-monotonic validation loss trajectory characteristic of fine-tuning a pretrained backbone on a small target dataset. The optimized three-phase freeze-unfreeze strategy partially addresses this instability by decoupling head training from backbone fine-tuning and applying a lower learning rate to the backbone parameters, but the validation accuracy still exhibited oscillations during the layer3-unfreezing phase (Phase 3, epochs 19-30), dropping from 81.24% at epoch 21 to 75.45% at epoch 24 before recovering to 83.61% at epoch 29.

The audio sub-network, in contrast, trained stably across all runs, with monotonically increasing validation accuracy in the primary run from epoch 1 (12.47%) to convergence, before the EarlyStopping callback halted training. The much higher initial validation loss values (7.9 at epoch 1) relative to those of the fusion model (0.88 at epoch 1) reflect the substantially more challenging nature of single-modality emotion discrimination from the 40-dimensional MFCC feature space compared to the 568-dimensional fused representation. The fusion model converged exceptionally rapidly in the primary run, reaching 91.86% validation accuracy within

the first epoch and 100% validation accuracy by epoch 7, which is indicative of the high discriminative capacity of the 568-dimensional representation relative to the complexity of the 8-class classification task.

The construction of the 568-dimensional fusion feature matrix over the full 2,452-sample corpus required approximately 4 hours and 10 minutes in the primary run (executed on a local GPU), corresponding to approximately 6.12 seconds per sample, due to the sequential per-sample video inference pipeline that performed two separate forward passes per sample (extract_512 and direct forward pass). This latency was reduced to 16 seconds total in the optimized run by replacing per-sample inference with a fully batched forward pass using the return_feat=True flag of the video model, representing a reduction of approximately three orders of magnitude and constituting one of the most significant engineering improvements introduced in the optimized configuration.

4.6.3 Methodological Limitations and Validity Considerations

As explicitly documented in the project appendix, the primary experimental protocol harbors three methodological limitations that inflate the reported performance relative to what would be observed in a rigorous actor-independent evaluation. First, in the primary run, StandardScaler statistics and SMOTE interpolation are fitted on the full corpus prior to splitting, creating a subtle leakage pathway through which test-set statistical properties influence the training-set preprocessing. Second, the video checkpoint in the primary run is selected on the basis of test-set accuracy at each epoch, meaning that the reported test-set accuracy represents the maximum accuracy observed on the test set during training rather than a prospective evaluation on a held-out set. Third, the random-split protocol places the same face and voice on both sides of the train-test boundary, allowing the model to exploit speaker-identity cues rather than genuine emotion-discriminative features.

These three limitations, collectively, account for the observed gap between the primary run fusion accuracy (94.50%) and the actor-independent fusion accuracy (94.13%), as well as the substantially larger gap in the audio modality (80.73% vs. 65.25%) and video modality (93.48% vs. 83.61%). The actor-independent results are recommended for citation when reporting MindSync performance in contexts requiring rigorous generalization claims, as they represent a more conservative and defensible estimate of the system's ability to detect depression-related affective indicators in recordings from speakers not encountered during training.

4.7 Chapter Summary

Table 4.12 provides the final consolidated metric summary for all three MindSync sub-systems across both experimental protocols, as extracted from the FULL METRIC SUMMARY (which reports both macro and weighted metrics, AUC-ROC, and PR-AUC for the optimized tuned run) and from the primary classification reports.

Table 4.12: Final Consolidated Metric Summary — MindSync All Sub-Systems

Model / Run	Acc. (%)	Macro F1 (%)	Wtd F1 (%)	Macro P (%)	Macro R (%)	AUC-ROC (%)
Audio — Actor-Indep.	65.25	64.25	65.25	65.10	65.47	—
Video — Actor-Indep.	83.61	83.48	83.94	83.63	84.75	—
Fusion — Actor-Indep.	94.13	93.60	94.18	94.22	93.97	—
Audio — Primary Run	80.73	80.31	79.65	81.00	81.00	—
Video — Primary Run	93.48	93.27	93.78	93.21	93.37	—
Fusion — Primary Run	94.50	93.87	94.22	95.22	93.32	—
Audio — Tuned Run	85.55	85.51	85.48	85.65	85.58	98.60
Video — Tuned Run	96.54	96.30	96.54	96.46	96.24	99.46
Fusion — Tuned Run	96.74	96.46	96.74	96.75	96.25	99.38

The experimental evaluation presented in this chapter demonstrates that the MindSync multimodal framework achieves consistent, substantial improvements over either unimodal sub-network across all evaluation conditions and experimental protocols. The fusion model reaches 96.74% accuracy in the tuned run and 94.13% in the actor-independent evaluation, with F1-scores for the depression-relevant sad, neutral, and fearful categories consistently above 89% in the actor-independent condition. These results validate the core design hypothesis underlying the MindSync architecture: that feature-level fusion of complementary audio and video representations, mediated by a 568-dimensional hybrid concatenation and a learned MLP classifier, yields a richer and more generalizable affective representation than either modality can provide independently, and provides a technically sound computational foundation for the rule-based Mental Health Inference Engine described in Chapter 5.

Chapter 5

Conclusion and Future Work

5.1 Introduction

The present chapter synthesizes these contributions into a coherent overall assessment of what the MindSync system has achieved, where its current boundaries lie, and how future work may extend its applicability. Section 5.2 summarizes the key technical contributions delivered across the preceding chapters. Section 5.3 distills the most substantive scientific and practical insights derived from the experimental results of Chapter 4. Section 5.4 provides an objective and transparent assessment of the limitations that bound the current implementation. Section 5.5 presents a structured set of future development directions and recommendations — including the expansion toward multilingual and cross-cultural deployment — intended to advance the system toward greater clinical applicability and scientific rigor.

5.2 Summary of Contributions

This thesis has presented MindSync, a feature-level multimodal deep learning framework for automated emotion recognition and depression-indicator detection from synchronized audio-visual recordings. The system was designed, trained, and evaluated on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [29], and its architecture encompasses three independently trained yet functionally integrated components, each constituting a distinct technical contribution.

5.2.1 Audio Sub-Network

The audio sub-network is a hierarchical sequential architecture comprising four one-dimensional convolutional blocks (64, 128, 256, and 512 filters), two stacked Bidirectional Long Short-Term Memory layers (256 and 512 units respectively), and a Bidirectional SimpleRNN layer of 512 units, operating on 40-dimensional Mel-Frequency Cepstral Coefficient (MFCC) feature vectors. Class imbalance was addressed through the Synthetic Minority Over-sampling Technique (SMOTE), and a systematic grid search over 18 hyperparameter configurations identified the optimal training setting, yielding a tuned validation accuracy of 85.03% and a test accuracy of 85.55% in the optimized experimental run.

5.2.2 Video Sub-Network

The video sub-network is built upon an R3D-18 three-dimensional convolutional backbone pretrained on the large-scale Kinetics-400 action recognition dataset, augmented with three auxiliary task-specific 3D convolutional refinement layers (256, 128, and 64 filters). A three-phase progressive freeze-unfreeze fine-tuning schedule was designed and applied, in which the classification head, the layer4 block, and the layer3 block of the backbone were successively unfrozen across 30 training epochs, enabling systematic and stable domain adaptation from action recognition to facial emotion classification. A complementary two-stage hyperparameter search over 12 video configurations — comprising a 10-epoch screening phase followed by a 25-epoch refit of

the top three configurations — yielded a best validation accuracy of 95.17% and a test accuracy of 96.54% in the optimized run.

5.2.3 Feature-Level Fusion Architecture and MHIE

The feature-level fusion mechanism constructs a 568-dimensional hybrid representation by concatenating four constituent vectors: the 40-dimensional MFCC descriptor, the 8-dimensional audio Softmax probability distribution, the 512-dimensional R3D spatiotemporal embedding, and the 8-dimensional video Softmax probability distribution. This representation is processed by a five-layer Multi-Layer Perceptron (MLP) classifier — with hidden layers of 256, 128, 64, and 32 units — optimized using AdamW with weight decay and StepLR scheduling. The fusion system achieved a peak test accuracy of 96.74% in the tuned experimental run and 94.13% under the more rigorous actor-independent evaluation protocol. The Mental Health Inference Engine (MHIE) extends the system beyond emotion classification by applying four transparent rule-based scoring functions that map the fusion Softmax output onto clinically motivated affective indicator scores for depression, anxiety, stress, and emotional blunting, providing an interpretable clinical interface without requiring any supplementary training data.

5.3 Key Findings and Insights

The experimental evaluation conducted in Chapter 4 yielded several substantive scientific and methodological insights that extend beyond the specific numerical results of the MindSync system.

Feature-level multimodal fusion provides consistent and robust gains over unimodal baselines. Across all evaluation conditions and experimental protocols, the fusion classifier outperformed both the audio-only and video-only sub-networks by substantial margins. In the primary run, the fusion gain over the audio baseline was 13.77 percentage points; in the optimized actor-independent run, the corresponding gain was 28.88 percentage points. The consistency of this pattern across two protocols with materially different partitioning strategies confirms that the improvement is attributable to genuine cross-modal complementarity rather than to protocol-specific artifacts.

The video modality provides substantially more discriminative features than the audio modality under controlled recording conditions. The video sub-network outperformed the audio sub-network by approximately 12–15 percentage points in the primary run and by approximately 18 percentage points in the actor-independent run. This differential reflects the greater individual distinctiveness of facial expression configurations relative to vocal acoustic properties within the RAVDESS corpus, and the higher representational capacity of the Kinetics-400 pretrained spatiotemporal backbone compared to the 40-dimensional MFCC feature space.

Actor-independent evaluation reveals a substantially more conservative performance profile than random-split evaluation. The gap between the random-split audio accuracy (80.73%) and the actor-independent audio accuracy (65.25%) — a difference of 15.48 percentage points — quantifies the magnitude of the speaker-identity shortcut inadvertently available in random-split evaluations of RAVDESS. This finding underscores the importance of actor-disjoint partitioning for any system intended to generalize to previously unseen speakers and is consistent with findings reported in the literature [30].

Structured hyperparameter optimization yields material performance improvements even for well-studied architectures. The two-stage grid search produced configurations that outperformed the reference default settings by 5.31 percentage points for the audio modality (80.73% → 85.55%) and approximately 2.82 percentage points for the video modality (93.48% → 96.54%), demonstrating the practical value of systematic configuration search over manual tuning, particularly when the training corpus is small and validation accuracy estimates are consequently noisy.

Low-arousal emotion categories present the most persistent classification challenge. Across all modalities and both experimental protocols, the neutral, calm, and sad categories consistently exhibited the lowest per-class F1-scores relative to high-arousal categories such as angry, happy, and disgusted. This finding is directly relevant to the depression-indicator detection application of MindSync: the three emotion categories most closely associated with depressive affect in the clinical literature are precisely those that the system finds most challenging to discriminate, motivating the incorporation of additional modalities and clinical data sources in future development.

5.4 Limitations

Notwithstanding the performance results reported in Chapter 4, the MindSync system is subject to several substantive limitations that must be acknowledged before any clinical or operational deployment is considered.

Dataset domain gap. The RAVDESS corpus consists of controlled recordings of acted emotional expressions delivered by 24 professional actors under laboratory conditions. The acoustic and visual properties of acted emotional speech differ substantially from those of spontaneous expression characteristic of genuine clinical populations, particularly for low-arousal categories most relevant to depression detection. Models trained exclusively on acted corpora are known to generalize poorly to naturalistic or clinical settings [30], and MindSync is not exempt from this limitation.

Limited corpus scale. The full RAVDESS corpus comprises 2,452 paired audio-visual samples, a comparatively small training set for a system comprising three independently parameterized deep learning components. Reliance on SMOTE interpolation and transfer learning from Kinetics-400 partially mitigates this constraint, but does not eliminate the risk of overfitting to corpus-specific speaker or recording characteristics.

Absence of clinical ground truth. The MHIE scoring functions are rule-based heuristics motivated by the clinical literature but not validated against any reference standard of depression severity, such as clinician-administered PHQ-9 or HAM-D ratings. The affective indicator scores produced represent exploratory research outputs rather than clinically validated diagnostic signals, and must not be used as the basis for clinical decision-making without prior prospective validation against a labeled clinical cohort.

Single-session temporal snapshot. MindSync processes a single short-duration audio-visual recording, producing a momentary affective assessment. Major Depressive Disorder is inherently a temporally extended condition characterized by persistent patterns rather than episodic affect, and a single session captures only a

snapshot of a complex dynamic affective trajectory. Longitudinal deployment across multiple sessions would be required to produce clinically meaningful trend information.

Evaluation restricted to eight discrete emotion categories. The RAVDESS taxonomy provides eight categorical emotion labels that do not capture the continuous, multidimensional nature of affect as conceptualized in dimensional models such as the valence-arousal space [21]. Depression-related affective states such as anhedonia, psychomotor retardation, and blunted affect are not directly represented as discrete categories in the RAVDESS label set, limiting the direct applicability of the classification output to clinical depression screening without the interpretive mediation of the MHIE.

5.5 Future Work and Recommendations

The limitations identified in Section 5.3, together with the analytical insights derived from the experimental evaluation in Chapter 4, motivate a structured set of future development directions that would substantially advance the clinical applicability and scientific rigor of the MindSync framework.

Domain adaptation to naturalistic clinical datasets. The most consequential near-term extension of MindSync is fine-tuning and re-evaluation on clinically relevant corpora that provide naturalistic, spontaneous emotional expression data with concurrent depression severity ratings. The DAIC-WOZ [6] and Extended DAIC (E-DAIC) corpora, annotated with PHQ-8 scores, and the AVEC challenge datasets [21], which include continuous affect annotations, represent the most directly applicable target domains. Domain adaptation techniques including adversarial training, domain-invariant feature learning, and progressive fine-tuning on small clinical target sets should be investigated to bridge the domain gap identified in Section 5.3.

Integration of additional modalities. The current MindSync architecture processes audio and video signals exclusively. Substantial performance improvements could be achieved by incorporating electroencephalography (EEG), which provides direct neural correlates of emotional state [15]; galvanic skin response (GSR) and other physiological signals that capture autonomic nervous system responses associated with emotional arousal; and transcribed verbal content, which encodes semantic and pragmatic emotional cues inaccessible to acoustic and visual processing.

Longitudinal deployment and temporal aggregation. Future implementations should incorporate a longitudinal deployment module that aggregates MHIE indicator scores across multiple recording sessions and applies time-series analysis methods — including moving averages, trend detection, and change-point identification — to identify persistent affective patterns indicative of clinical depression rather than transient emotional states.

Data-driven MHIE scoring calibration. The current rule-based scoring functions employ fixed expert-determined weighting coefficients that have not been learned from labeled clinical data. Future work should replace or augment these functions with data-driven mechanisms trained on corpora providing direct behavioral correlates of depression severity. Logistic regression, gradient boosting, or shallow neural network classifiers trained on the fusion Softmax output as input features and PHQ-9 or HAM-D severity ratings as targets represent natural candidates for this calibration task.

Explainability and clinical interpretability. For deployment in a clinical screening context, transparent and interpretable predictions are a prerequisite for regulatory approval and clinical acceptance. Future iterations of MindSync should incorporate gradient-based feature attribution methods such as SHAP [47] or Grad-CAM applied to the spatiotemporal video activations, enabling clinicians to inspect which temporal frames and facial regions most strongly influenced the system's predictions. Attention weight visualization for the audio BiLSTM layers would similarly provide interpretable indicators of which temporal segments of the speech signal carried the most diagnostic information.

Transformer-based and self-supervised backbone replacement. Recent advances in self-supervised speech representation learning — including wav2vec 2.0 [39] and HuBERT — have demonstrated substantially superior performance on downstream emotion recognition tasks by learning latent representations directly from raw waveforms without label supervision. Replacing the MFCC front-end with a pretrained self-supervised audio encoder would be expected to provide richer and more generalizable acoustic features, particularly for low-arousal emotion categories where MFCC discriminability is limited. Similarly, replacing the R3D-18 video backbone with a Vision Transformer (ViT) [42] or a video-adapted transformer architecture such as TimeSformer would enable the capture of long-range spatiotemporal dependencies beyond the local receptive fields of convolutional kernels.

In summary, the MindSync framework demonstrates that feature-level multimodal fusion of complementary audio and video representations — mediated by a 568-dimensional hybrid concatenation and a learned MLP classifier — provides a robust, generalizable, and interpretable foundation for automated emotion recognition and depression-indicator detection from audio-visual recordings. The system achieves consistent performance improvements over unimodal baselines across multiple evaluation protocols, reproduces the benchmarks of the reference companion study [30], and provides a clinically interpretable affective indicator scoring layer through the MHIE. The limitations and future directions identified in this chapter define a concrete research agenda whose pursuit would advance MindSync from a validated research prototype toward a system of genuine clinical utility in automated depression screening applications.

Multilingual deployment and cross-lingual generalization. The current MindSync implementation is realized as a functional web application that processes audio-visual input exclusively in English, which constrains its applicability to a linguistically narrow deployment context despite the language-independence of many of the underlying visual and paralinguistic cues it exploits. Extending the platform into a multilingual web application capable of processing and analyzing speech across diverse languages represents a natural and clinically significant next step, as it would substantially broaden the system's real-world reach and support screening in linguistically diverse populations that current English-only tools cannot serve. Realizing this extension, however, introduces non-trivial technical challenges. Foremost among these is the need for cross-lingual transfer learning, whereby acoustic emotion cues learned from the English-centric RAVDESS corpus must generalize to languages with markedly different prosodic, phonetic, and rhythmic structures without requiring costly re-annotation of emotion-labeled speech for every target language. Addressing this would likely involve replacing the current MFCC-based audio front-end with a multilingual self-supervised speech encoder — such as a cross-lingual variant of wav2vec 2.0 (XLS-R) [39] — pretrained on large multilingual speech corpora, so as

to obtain language-agnostic representations that capture paralinguistic emotional markers (pitch contour, energy, spectral tilt) independently of the phonemic inventory of the input language. In parallel, the video sub-network, being largely language-independent, would require comparatively minor adaptation, reinforcing the expectation that the video modality could serve as a stabilizing anchor for fusion performance during early-stage multilingual rollout, while the audio and text-adjacent components mature through targeted domain adaptation. Beyond the modeling layer, front-end internationalization (i18n) of the web application interface would also be required to support multilingual user interaction and reporting. Collectively, these developments would improve the reproducibility and cross-cultural generalizability of MindSync's affective indicators and position the system for meaningful international deployment as an accessible, language-inclusive screening tool.

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ملخص المشروع

يُعد الاضطراب الاكتئابي الجسيم (MDD) واحدًا من أبرز أسباب الإعاقة على مستوى العالم، إذ يؤثر على أكثر من 280 مليون شخص، ويفرض أعباءً اجتماعية واقتصادية كبيرة على أنظمة الرعاية الصحية. يعتمد التقييم الحالي للاكتئاب بشكل أساسي على أدوات سريرية ذاتية مثل استبيان صحة المريض-9 (PHQ-9) ومقياس هاملتون لتقييم الاكتئاب (HAM-D)، وهي أدوات تواجه تحديات التحيز في الإبلاغ ومحدودية قابليتها للتوسع. وعلى الرغم من الإمكانيات الواعدة للأساليب الآلية، فإن كثيرًا من الأنظمة القائمة تعتمد على نمط واحد فقط من البيانات، سواء الصوت أو تعبيرات الوجه، مما قد يُغفل معلومات وجدانية تكاملية مهمة.

تقدّم هذه الدراسة نظام MindSync، وهو إطار عمل للتعلم العميق متعدد الوسائط على مستوى السمات، مصمم للكشف الآلي عن مؤشرات الاكتئاب من تسجيلات صوتية-مرئية متزامنة. يدمج الإطار المقترح شبكتين فرعيتين مُدرّبتين بشكل مستقل: شبكة للصوت وأخرى للفيديو. تعالج شبكة الصوت 40 معامل MFCC باستخدام أربع كتل تلافيفية أحادية البعد، وطبقتي BiLSTM ثنائيي الاتجاه، وطبقة SimpleRNN ثنائية الاتجاه. أما شبكة الفيديو فتستخدم نموذج R3D-18 التلافيفي ثلاثي الأبعاد المُدرّب مسبقًا على مجموعة بيانات Kinetics-400، مع إضافة طبقات Conv3D لاستخلاص السمات المكانية-الزمانية الخاصة بالوجه. تُدمج التمثيلات الناتجة من كل نمط مع توزيعات احتمالات Softmax الخاصة بها لتكوين متجه سمات هجين بأبعاد 568، يُصنّف عبر شبكة دمج من نوع MLP تم تحسينها باستخدام AdamW وجدولة StepLR.

يُدرّب نظام MindSync ويُقيّم على مجموعة بيانات RAVDESS، التي تضم 2,452 تسجيلًا صوتيًا-مرئيًا متزامنًا موزعة على ثماني فئات انفعالية أداها 24 ممثلًا محترفًا. ولمعالجة عدم توازن الفئات، طُبِّقت تقنية SMOTE حصريًا على قسم التدريب ضمن حيز سمات MFCC، لتفادي تسرب البيانات إلى مجموعة التقييم.

يقوم محرك استدلال شفاف قائم على القواعد، هو محرك الاستدلال الصحي النفسي (MHIE)، بربط الاحتمالات الانفعالية الثمانية بأربعة مؤشرات وجدانية: احتمال الاكتئاب، القلق، التوتر، والتبذل الانفعالي، ويُمثل ذلك تحليلًا لأنماط انفعالية وليس تشخيصًا سريريًا.

أظهرت النتائج التجريبية دقة بلغت 85.55% و96.54% و96.74% لأنظمة الصوت فقط، والفيديو فقط، والنظام متعدد الوسائط على التوالي، مع قيم AUC-ROC ماكرو مقابلة بلغت 98.60% و99.46% و99.38%. ويؤكد الأداء المتفوق للنظام متعدد الوسائط أن دمج التمثيلات الصوتية-المرئية التكاملية يوفر تمثيلًا وجدانيًا أكثر ثراءً وتمييزًا مقارنة بأي نمط منفرد. وسيركز العمل المستقبلي على التكيف مع مجالات بيانات سريرية مثل DAIC-WOZ، ودمج الإشارات الفسيولوجية، والتحقق المستقبلي مقابل مقاييس شدة الاكتئاب التي يديرها الأطباء.



الجهة المانحة



دعم الصحة النفسية باستخدام الذكاء الاصطناعي متعدد الوسائط من خلال تحليل تعبيرات الوجه والإشارات الصوتية.

رسالة مقدمه من

بثينة رمضان كمال

زهراء مصطفى عليوة

بسمه فتوح طلحه

رحاب معوض محمد

خلود محمود محمد

رسالة مقدمة استكمالاً لمتطلبات الحصول على

دبلومة Digilians 9 Months
مسار الذكاء الاصطناعي التطبيقي و تحليل البيانات

يعتمد من لجنة الممتحنين

التوقيع

الإسم



وزارة الاتصالات
وتكنولوجيا المعلومات



الجهة المانحة



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رسالة مقدمة استكمالاً لمتطلبات الحصول على دبلومة Digilians لمدة تسعة أشهر في الذكاء الاصطناعي التطبيقي وتحليل البيانات

تحت إشراف

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أغسطس 2026